

Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

How much can the training dynamics of a deep network be compressed? Compressing a trained function does not by itself compress its learning: networks that agree on every input can diverge once training resumes. We retain depth, effective nonlinearity and feature learning, and replace each neuron’s response history by finitely many evolving Legendre moments in a clock driven by residual RMS. The resulting autonomous dynamics reconstruct the learned interactions while retaining the initialized matrices exactly. For fixed depth and training data, canonical Gaussian initialization, zero initial readout and sufficiently small fixed labels, we prove exponential fitting at every memory order and approximation through the entire training trajectory. Against the dense population predictor, the whole-input test error is at most $Cq^{-2} \exp\{K\sqrt{\log(e+q)}\}$ plus a width remainder that vanishes in probability; the constants are independent of width and time. Every diverging order $q = o(n)$ therefore reaches the same population learning process with $O(Lmnq)$ evolving history coordinates instead of $O(Ln^2)$ learned weights. Broader-label finite-time guarantees and an exact hidden-learning-rate rescaling describe extensions of the central regime. Circle, sphere and MNIST experiments show that small memory orders reproduce dense fitted predictions on unseen inputs, while the tested frozen-kernel and rank-matched factor baselines can select substantially different functions. The result isolates a compressible part of deep learning: the response memory that carries its evolving interactions.

1 Introduction

A missing model of learning. The Turing machine is the canonical model of computation. It is idealized—it has no fixed bound on memory—and it is simple enough to reason about, while physical computers, from the von Neumann architecture onward, are understood as its finite-resource implementations. One might view neural networks as playing the role in learning that physical computers play in computation; what this analogy still lacks is a counterpart to the Turing machine—a canonical idealized model of learning. Deep neural networks trained by gradient descent are the obvious candidates for implementations of such a model, but the object they would implement is not yet known. The obstacle is that their training dynamics are enormously complex and have no simple representation.

The limit, and why to compress it. A natural place to look for the idealized object is the limit of infinite width. Under feature-learning scalings of the type used in tensor-program and mean-field theory (Yang and Hu, 2021; Nguyen and Pham, 2023), width-dependent learning rates preserve order-one feature evolution, and suitable population observables obey deterministic limiting laws. The limit is the network with no resource constraint, as the Turing machine is the computer with none; a finite network would then be this limit plus a quantified deviation from it, as in a central limit theorem. Going to infinite width to simplify seems paradoxical, since the dynamical system grows with width. Statistical physics resolves the paradox in other settings: a gas of 10^{23} molecules

is simpler to describe than a gas of ten, because its macroscopic behaviour is captured by a few collective variables. Training presents two sources of complexity: the number of neurons, and the response history introduced when evolving weights are eliminated from the description. The weights themselves are a Markovian state; response-based limits express their learned interactions through the past. Mean-field and tensor-program limits address width. Whether the response history can also be *compressed* is our question. We do not seek compression for its own sake: a compressed description must retain what drives learning and may discard the rest, so the state that survives is a candidate for the variables that govern learning, and a compressed system that reproduces training would be a step toward the idealized model.

What the model must keep. Several influential simplifications isolate different aspects of training. Deep linear networks admit exact solutions but remove the nonlinearity (Saxe et al., 2014); one-hidden-layer mean-field models remove multilayer hidden interactions (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); and the neural tangent kernel keeps depth and nonlinearity but freezes the features (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple precisely because of what it gives up. Our target keeps all three: *depth* (more than one hidden layer), *nonlinearity* that stays effective during training and in the limit, and *feature learning* (hidden representations move at leading order). It must also reproduce the whole trajectory, not a snapshot: two networks that compute the same function on every input can have different Jacobians and backward responses, and once training resumes the discrepancy accumulates. Together with a specified population limit, this contract fixes the object we seek to understand.

The gap, and one axis for every description. Width compression leaves a further resource question. Tensor Programs carry variables indexed by every earlier training step (Yang and Hu, 2021), and feature-learning dynamical mean-field theory (DMFT) evolves two-time kernels whose memory grows quadratically with the number of time points (Bordelon and Pehlevan, 2022). These descriptions encode learned interactions through temporal responses rather than explicit learned matrices, a substantial reduction in width dependence, but their state grows with the number of retained time points. We ask whether evolving learned interactions can instead be carried by a fixed number of neuron-level coordinates, with a bound on the error of the complete training trajectory. The question suggests one axis on which every description of training can be placed: fix a horizon and an accuracy, and ask how much *evolving* state a description needs to reproduce the trajectory to that accuracy, and whether that state grows with the number of elapsed steps. Descriptions that freeze the features, drop depth or drop the nonlinearity target a different training regime, and enlarging their state alone supplies no general approximation guarantee for ours. Those that keep the contract each pay for the limit in a different currency: the dense network in neurons, Tensor Programs and DMFT in stored history, the Neural Tangent Hierarchy in tensors indexed by tuples of training samples. Response memory pays in an approximation order: no coordinates are added merely because more steps elapse. Under the small-label conditions proved below, one order controls the entire physical-time interval; more generally the required order can depend on the horizon and the task. Definition 2.1 makes the axis precise, Table 1 (Section 5) places the descriptions on it once the results are in hand, and Appendix F works out, under stated hypotheses, what each would cost at a prescribed accuracy.

Key contribution. The learned matrix is a paired memory of forward and backward responses. We compress that memory into an autonomous dynamics and control its feedback on the signals it records. Our central theorem makes this statement uniform over the whole learning process: in a

small-label Gaussian regime, every memory order fits exponentially, and order q tracks the dense population predictor with error $q^{-2+o(1)}$ plus a vanishing width remainder (Theorem 3.1). The regime permits any fixed depth and a broad class of nonlinear activations; labels are small but fixed as width grows, and hidden learning retains its canonical scaling. As a consequence, any diverging order $q = o(n)$ converges to the same limiting predictor while using a vanishing fraction of the dense model’s evolving state (Corollary 3.2). This is compression of an entire learning trajectory, not a factorization of its endpoint. Small orders reproduce the fitted function in experiments where frozen kernels and rank-matched trained factors do not (Section 4).

2 Setting

Fix m training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, width n and $L \geq 2$ hidden layers:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} x_a / \sqrt{d}, & h_a^{(1)} &= \phi(z_a^{(1)}), \\ z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi(z_a^{(\ell)}), \quad 2 \leq \ell \leq L, \\ f_a &= \frac{w^\top h_a^{(L)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}. \end{aligned} \tag{1}$$

The activation ϕ may differ between layers; the loss is $\mathcal{L} = \rho^2$. The backward responses, which exclude the residual, are $\delta_a^{(L)} = w \odot \phi'(z_a^{(L)})$ and $\delta_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}$. With block mobilities $(n, 1, \dots, 1, n)$, gradient flow reads

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} \frac{x_a^\top}{\sqrt{d}}, \quad \dot{W}^{(\ell)} = -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top \quad (2 \leq \ell \leq L), \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \tag{2}$$

We write $\theta = (W^{(1)}, \dots, W^{(L)}, w)$ for all parameters and $\dot{\theta} = F(\theta)$ for (2). Hidden entries are initialized as $\mathcal{N}(0, 1/n)$, first-layer entries as $\mathcal{N}(0, 1)$, and the stored readout is specified by the result being applied; the central theorem uses $w(0) = 0$ exactly.

For comparisons across widths we use the normalized parameter distance

$$d_n(\hat{\theta}, \theta) = \frac{\|\widehat{W}^{(1)} - W^{(1)}\|_F}{\sqrt{n}} + \sum_{\ell=2}^L \|\widehat{W}^{(\ell)} - W^{(\ell)}\|_F + \frac{\|\hat{w} - w\|_2}{\sqrt{n}}. \tag{3}$$

These scalings agree with the block mobilities and with order-one forward predictions.

Population flow. As $n \rightarrow \infty$, the neurons of layer ℓ become a probability space Ω_ℓ with expectation $\mathbb{E}_\ell$: hidden coordinates $z^{(\ell)}, h^{(\ell)}$ become random variables in $L^2(\Omega_\ell)$, the first-layer weights a random row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^d)$, the readout $w \in L^2(\Omega_L)$, and each hidden matrix a bounded operator $W^{(\ell)} : L^2(\Omega_{\ell-1}) \rightarrow L^2(\Omega_\ell)$ used together with its true adjoint. The same equations (2) hold with $\frac{1}{n} \sum_i$ replaced by $\mathbb{E}_\ell$ and outer products $\delta h^\top / n$ by the rank-one operators $g \mapsto \delta \mathbb{E}[h g]$. Such limits are established in several scopes—for fixed depth on a short horizon, and globally in special models—with finite-width gradient flow converging to them on compact horizons; a global theorem for general data and depth is open. In the small-label regime of Theorem 3.1, the Gaussian construction and the finite-activity estimate identify the dense population predictor for all physical time. The proof is included in Appendix A.

What counts as compression. An exact compression reproduces the reference trajectory; an approximate one reproduces it up to a controllable error. A compression may also match the current state and velocity while its small errors compound until the trajectories separate; we ask for the stronger, global notion.

Definition 2.1 (Global approximation on compact horizons). Autonomous systems indexed by an order q , with trajectories θ_q , *globally approximate* a reference flow θ if for every $T < \infty$

$$\sup_{0 \leq t \leq T} \|\theta_q(t) - \theta(t)\| \longrightarrow 0 \quad (q \rightarrow \infty).$$

The rate in q quantifies the trade-off between the size of the compressed state and its accuracy. We call the approximation *all-time* when the same conclusion holds with the supremum over $t \geq 0$. Our central result establishes this stronger notion, with an additional vanishing width remainder when the target is the population flow.

The size in question is the *moving state* of a description: the numbers it updates while computing training, excluding fixed sources such as the initialization or a frozen kernel.

3 Closing the training dynamics: exact and approximate

3.1 Exact closure: operators and an obstruction

A closed description of training evolves a state of one of three kinds: finitely many *scalars*, independent of width and training time; finitely many *populations* (functions, random variables or laws on fixed spaces, such as the law of a neuron’s weights), whose particle implementation stores one value per particle and field; or *operators* between population spaces, the limits of matrices, which record how each row relates to each column, a coupling that one-neuron marginal statistics discard. A finite number of named fields or operators need not contain finitely many scalars, so every claim below names both the type and the size of the retained state.

Before allowing errors we ask what exact capture of the population flow can look like. A natural candidate without growing history is an operator ODE: (2) read with the hidden matrices $W^{(\ell)}$ as operators. In a deep linear benchmark—one input, three linear hidden layers, order-one initialization—this can be proved completely: the learned part of the whole network becomes a single trace-class operator that evolves by an explicit autonomous cubic equation driven by the residual, and finite gradient flow and gradient descent converge to it on every compact horizon in the stated limits (Appendix E). But one named operator can contain infinitely many degrees of freedom, so an operator equation alone supplies no finite state–accuracy trade-off; such a trade-off requires a further approximation of the operator. At the other extreme, in the same linear benchmark no encoder formed from a fixed finite list of tensor contractions of the weights with width-independent bounded degree admits a polynomial autonomous scalar ODE and polynomial readout that reproduce the output dynamics on an open set of states at all large widths (Appendix E). The reason is that repeated differentiation of the output forces ever larger connected contractions: the $(2j - 1)$ st derivative contains $\frac{1}{n} (W^{(1)})^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j W^{(1)}$, of degree $4j + 2$, which no bounded-degree alphabet can express. The result is restricted to this encoder and readout class; it identifies a concrete obstacle already present with linear features, not a no-go theorem for all finite scalar descriptions. Between the two, whether the exact operator can be compressed into finitely many populations is open: rank-one gradient updates do not by themselves imply a fixed rank bound on their time integral. The main result takes a constructive route: accepting controlled error gives an explicit autonomous system with a quantitative state–accuracy trade-off.

3.2 Approximate closure: the idea

Integrating (2), every hidden matrix is

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \quad (4)$$

The learned part is an integral of backward against forward responses. Two direct representations are the dense matrix and the accumulated response history. A familiar analogy is a moving ball: under a known force law, its current position and velocity can determine the future without storing its past positions. Here we seek history coordinates that retain enough information for the network’s own law of motion. Regular response histories suggest that a few coefficients can summarize the interaction integral. Two things must hold: the summary must be *closed*, so that it evolves autonomously, and *accurate* over the whole of training, which alternates between bursts of learning and plateaus.

3.3 Closure: shifted Legendre moments

Let p_k be the shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and $\int_0^1 p_j p_k = \mathbf{1}_{j=k}/(2k+1)$. Their derivatives reduce to lower modes,

$$x p'_k(x) = k p_k(x) + \sum_{j < k} (2j+1) p_j(x), \quad (5)$$

and this is what makes the moment system close exactly. The identity underlies online polynomial memory in HiPPO (Gu et al., 2020); the new element is that the memorized signals are the network’s own forward and backward responses, coupled back through the reconstructed network. The first moments of a history are its mean, its linear trend (the first polynomial rises from -1 at the start to 1 now) and its quadratic shape; each added moment corrects what the previous ones miss (Figure 8, Appendix I).

3.4 Accuracy: a clock that follows learning

Concentrated variation can require many polynomial modes in physical time. We therefore measure history in a clock τ that runs at the speed of learning,

$$\dot{\tau} = \rho, \quad \tau(0) = 1 : \quad (6)$$

fast when the residual is large and slow as it approaches zero. This speed is the square root of loss, not its time derivative; it need not stop at a positive-loss stationary point. The unit prefix $[0, 1]$ carries the initial forward responses and no backward response. For each hidden link $\ell \geq 2$ and sample a we store the first q Legendre moments, in the clock coordinate, of the forward response $h_a^{(\ell-1)}$ and of the residual-weighted backward response $r_a \delta_a^{(\ell)}/\rho$:

$$\bar{h}_{a,k}^{(\ell-1)} = \int_0^\tau h_a^{(\ell-1)} p_k(\xi/\tau) d\xi, \quad \bar{\delta}_{a,k}^{(\ell)} = \int_0^\tau \frac{r_a \delta_a^{(\ell)}}{\rho} p_k(\xi/\tau) d\xi, \quad 0 \leq k < q,$$

where ξ is the clock value at which each response was recorded. By (5) these moments obey the closed system

$$\begin{aligned} \dot{\bar{h}}_{a,k}^{(\ell-1)} &= \rho h_a^{(\ell-1)} - \frac{\rho}{\tau} \left[k \bar{h}_{a,k}^{(\ell-1)} + \sum_{j < k} (2j+1) \bar{h}_{a,j}^{(\ell-1)} \right], \\ \dot{\bar{\delta}}_{a,k}^{(\ell)} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k \bar{\delta}_{a,k}^{(\ell)} + \sum_{j < k} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \right], \end{aligned} \quad (7)$$

with $\bar{h}_{a,0}^{(\ell-1)}(0) = h_a^{(\ell-1)}(0)$ and all other moments zero, and the hidden matrices are reconstructed as

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{q-1} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top. \quad (8)$$

All responses in (7) are evaluated in the current *reconstructed* network, and the first layer and readout follow (2). The system is autonomous and never divides by ρ . Equivalently, with Π_q the $L^2(0, \tau)$ projection onto polynomials of degree below q ,

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau \left(\Pi_q \frac{r_a \delta_a^{(\ell)}}{\rho} \right) (\Pi_q h_a^{(\ell-1)})^\top d\xi, \quad (9)$$

and since cross terms vanish by orthogonality, the error in each matrix update is a *product* of a forward and a backward projection error. Figure 7(b) (Appendix I) illustrates this cancellation and distinguishes it from the separate stability argument that controls feedback.

The joint feature-learning clock. The learning-speed clock makes forward histories smooth but not every backward history. A second clock advances with the residual *plus* the speed of all forward and backward responses,

$$\dot{\tau} = \rho + \left\| \frac{d}{dt} (h_a^{(\ell-1)}, r_a \delta_a^{(\ell)} / \rho)_{\ell,a} \right\|_2, \quad \tau(0) = 1, \quad (10)$$

so that the whole response path moves at unit speed at most in the clock coordinate. History is placed by this clock but still weighted by the residual mass ρdt , so the Legendre modes are no longer orthogonal and one shared $q \times q$ Gram matrix is tracked alongside the moments; the reconstruction uses its inverse, the defect keeps the product form, and the clock speed can be evaluated explicitly by one directional forward and backward pass (Appendix D). The coordinate change is illustrated in Figure 4 (Appendix D).

3.5 A uniform approximation of the whole learning process

The central result concerns the learning-speed closure (7)–(8), with its own residual and responses. Its accuracy is measured against the function selected by dense training, including on inputs never used for training. For a test-input probability law μ , write

$$\mathcal{E}_\mu(g, f) = \left(\int \sup_{t \geq 0} |g(t, x) - f(t, x)|^2 d\mu(x) \right)^{1/2}. \quad (11)$$

The supremum is inside the integral: the estimate controls the entire trajectory at every test input, rather than only a selected stopping time.

Theorem 3.1 (All-time response-memory approximation). *Fix the depth $L \geq 2$, input dimension d , and m training inputs. In each layer let $\phi_\ell \in C^1(\mathbb{R})$ have a globally bounded derivative and a globally Lipschitz derivative. Use the independent Gaussian initialization of Section 2, with $w(0) = 0$, and the same initialization for dense and compressed training. Assume the limiting initial readout-feature Gram has a positive gap:*

$$\Gamma_{w,\infty}(0) := \lim_{n \rightarrow \infty} \frac{H^{(L)}(0)^\top H^{(L)}(0)}{mn} \succeq 2\lambda I_m, \quad \lambda > 0, \quad (12)$$

where the columns of $H^{(L)}$ are the training features. There is $Y_* > 0$, independent of width and order, such that for fixed labels with $Y = (m^{-1} \sum_a y_a^2)^{1/2} \leq Y_*$ the following hold.

There are constants $C, K, \kappa > 0$, independent of n, q, t , and common initialization events of probability tending to one, on which the dense flow and every closure order $q \geq 1$ exist for all time and satisfy

$$\rho_n(t), \hat{\rho}_{n,q}(t) \leq Y e^{-\kappa t}. \quad (13)$$

Their parameters converge as $t \rightarrow \infty$. Simultaneously for every order,

$$\sup_{t \geq 0} d_n(\hat{\theta}_{n,q}(t), \theta_n(t)) \leq C\omega(q) + b_n, \quad \omega(q) = \frac{\exp\{K\sqrt{\log(e+q)}\}}{q^2}, \quad b_n \xrightarrow{\text{Pr}} 0, \quad (14)$$

where b_n depends only on the dense reference, not on q .

The dense predictors converge to a deterministic population predictor f_∞ . For every fixed μ with $\int \|x\|^2 d\mu(x) < \infty$, there is an order-independent remainder $b_{n,\mu} \xrightarrow{\text{Pr}} 0$ such that, on the same events,

$$\mathcal{E}_\mu(\hat{f}_{n,q}, f_\infty) \leq C_\mu \omega(q) + b_{n,\mu}. \quad (15)$$

For every fixed bounded input set $\mathcal{X}$, the analogous bound holds for $\sup_{t \geq 0, x \in \mathcal{X}} |\hat{f}_{n,q}(t, x) - f_\infty(t, x)|$. Constants may depend on depth, data, activation bounds and the initial Gram gap. The label threshold is fixed as $n, q \rightarrow \infty$.

Thus the order error is $q^{-2+o(1)}$: every fixed exponent below two is available. The width remainder is a convergence statement, without an asserted numerical rate in n . For any square-integrable target function, the difference between the closure's and the dense population's test RMSE is at most (15), uniformly in time, by the reverse triangle inequality. In particular, a circle or sphere is a whole test domain covered by the theorem; a mesh is needed only to display its function.

The activation class includes tanh, softplus, GELU, SiLU and softsign; its values need not be bounded. The Gram condition is an initialization condition. One sufficient input criterion is nonzero, pairwise nonproportional inputs, with a nonpolynomial first activation and nonconstant subsequent activations in the stated class. The proof of this criterion and the theorem are in Appendix A.

Why the bound survives feedback and infinite time. Small labels first give a short total interval of learning activity. This keeps the initial feature-Gram gap open. The closure's exact velocity defect is the product of its forward and backward endpoint projection errors; their opposite powers of q cancel in a coarse bound relative to the residual. This proves exponential fitting for *every* order before using any approximation-to-dense argument. The sharper tracking estimate then uses the projection-energy identity to integrate the defect: its two history errors multiply. Gaussian tail control is needed only for the dense reference's backward carriers. Cutting those carriers in the proof and balancing the tail against feedback amplification produces the factor $\exp\{K\sqrt{\log(e+q)}\}$. The actual algorithm is unchanged.

Corollary 3.2 (Less evolving state for the same population flow). *Under Theorem 3.1, every deterministic sequence $q_n \rightarrow \infty$ converges to f_∞ in the all-time test metric, in probability. If also $q_n = o(n)$, then for fixed m, d, L this convergence uses a vanishing fraction of the dense network's evolving state:*

$$M_{\text{dense}} = (L-1)n^2 + n(d+1), \quad M_{\text{memory}} = 2(L-1)mnq_n + n(d+1) + O(1), \quad \frac{M_{\text{memory}}}{M_{\text{dense}}} \rightarrow 0. \quad (16)$$

Each learned hidden increment has rank at most mq_n and approximates the corresponding dense increment with error bounded by (14), uniformly over all time.

Proof. The order term and width remainder in (15) both vanish for every such sequence. The state count follows directly from the two sets of q_n length- n moments per sample and hidden link, together with the first matrix, readout and clock. Reconstruction (8) is a sum of mq_n outer products; its error is a block of the normalized parameter distance. $\square$

The order can therefore grow as slowly as $\lceil \log \log(e^e + n) \rceil$. No tuning of order against a hidden width-dependent amplification constant is required in this regime. At a fixed accuracy, one may first choose the order and then take width sufficiently large: the learned state is linear in width, whereas dense storage is quadratic. Quantifying the smallest sufficient width is a separate concentration question.

A broader finite-time branch. Small labels buy control through arbitrarily long training. Without that condition, arbitrary finite data with $Y \leq 1$ and bounded initial readout RMS admit a different result: on every fixed $[0, T]$, a width-independent C_T/q bound holds in a specified joint order-width region (Proposition B.1). For globally Lipschitz derivatives and zero readout, a sufficient threshold has the form $q \geq C_T \sqrt{n} \exp(C_T \sqrt{n})$. It requires neither a Gram gap nor successful fitting. Its quantifiers matter: the constant is uniform on that joint region, while the order threshold grows with width. It therefore extends the label range but does not supply the compression conclusion of Corollary 3.2. The general fixed-width guarantees, including $O_T(q^{-2})$ for the joint clock, remain in Appendices C and D.

Corollary 3.3 (Feature-learning strength). *Fix $0 < \alpha \leq 1$. Multiply the learning rates of every hidden weight, including $W^{(1)}$, by α^2 , keeping the readout rate unchanged. Use the learning-speed closure with $\dot{\tau} = \alpha\rho$, forward moment source $\alpha\phi h$, and backward moment source $\alpha^2 r\delta$; the dilation coefficient is $\alpha\rho/\tau$ and reconstruction is still (8). Then Theorem 3.1 applies whenever $\alpha Y \leq Y_*$, with the prediction bound divided by α and with the dense population trained by these same modified rates as reference.*

Proof. Run the canonical network on labels αy and denote its readout by $\tilde{w}$. Set $w = \tilde{w}/\alpha$, leaving hidden weights unchanged. Then $\tilde{f} = \alpha f$, $\tilde{r} = \alpha r$, and $\tilde{\delta} = \alpha\delta$. Every hidden update acquires the factor α^2 , while dividing the readout update by α restores its original rate. In (7), $\tilde{\rho} = \alpha\rho$ changes the clock and forward source, and $\tilde{r}\tilde{\delta} = \alpha^2 r\delta$ changes the backward source. The moments and hidden reconstruction are exactly those of the scaled-label canonical problem. Its output discrepancy divided by α is the discrepancy for the modified problem. This change of variables also transfers fitting and the state count. $\square$

For fixed positive α , hidden features still learn and the constants remain independent of width and time. This is an exact learning-rate rescaling, rather than a clipping rule or a theorem for unrestricted-label canonical training.

3.6 Cost and size at accuracy

Equation (16) counts the evolving model. The initialized hidden matrices are retained exactly, requiring another $(L-1)n^2$ fixed numbers. One right-hand-side evaluation costs

$$O(Lmn^2 + Lm^2nq + Lmnq + nmd). \quad (17)$$

The terms are, respectively, the forward/adjoint actions of W_0 , the learned low-rank actions on the batch, the moment updates using prefix sums, and the first layer. The joint clock additionally stores $O(q^2)$ numbers and costs $O(q^3 + Lmnq^2)$ for its Gram factorization and solves. These are

costs per evaluation; time discretization has its own accuracy requirement. Keeping W_0 explains why compressing the learned state does not itself reduce the dense matrix–vector runtime.

For the all-time theorem, an order contribution at most $\varepsilon/2$ is obtained with

$$q_\varepsilon = \varepsilon^{-1/2} \exp(O(\sqrt{\log(1/\varepsilon)})) = \varepsilon^{-1/2+o(1)}. \quad (18)$$

The moving history budget is consequently $O(Lmn\varepsilon^{-1/2+o(1)})$. For full population accuracy at confidence $1 - \delta$, let $N_\mu(\varepsilon, \delta)$ be a sufficiently large width that makes the width remainder at most $\varepsilon/2$ and the common event hold with that confidence. The theorem guarantees that this width is finite. Taking $n = N_\mu(\varepsilon, \delta)$ gives the sufficient history budget $O(LmN_\mu(\varepsilon, \delta)\varepsilon^{-1/2+o(1)})$ under exactly the theorem’s hypotheses. Quantifying N_μ is needed to turn this into an explicit power of ε . Appendix F isolates that step and the corresponding numerical obligations for history and hierarchy solvers; Corollary 3.2 already proves the reduction in moving state without a width-rate assumption.

3.7 What the population conclusion identifies

The result identifies one target function-valued learning process, $f_\infty(t, x)$, and a smaller family of evolving approximations to it. Every divergent order sequence converges to that target over the whole training interval $[0, \infty]$. At each fixed order, subsequential population predictor limits exist and inherit the pure order bound $C_\mu\omega(q)$; these limit predictors need not have been shown unique or deterministic. Identifying a unique autonomous population *moment-state* equation at fixed q is a finer question than the predictor approximation proved here.

The natural interpretation is a finite collection of memory fields for each sample and layer, coupled through the initialized operator. The learned action takes the form $g \mapsto -\frac{2}{m\tau} \sum_{a,k} (2k + 1) \bar{\delta}_{a,k}^{(\ell)} \mathbb{E}_{\ell-1}[\bar{h}_{a,k}^{(\ell-1)} g]$. The theorem establishes the accuracy and compression of its finite-width implementations without requiring that stronger identification.

4 Experiments

We integrate finite networks and compare response memory with dense gradient flow from the same initialization, using the learning-speed clock unless stated. The experiments probe the construction beyond the small-label theorem: their label sizes and, in the supplementary sphere task, nonsmooth ReLU activation need not satisfy its hypotheses. We first examine the developing function at shared physical times, then compare fitted functions at each model’s first crossing of training MSE 0.001, and then ask whether reproducing that function requires the response-memory dynamics. Shared-time discrepancies address trajectory tracking, while individually fitted endpoints test whether the closure reproduces the function selected by dense training. The trajectory panel uses a fresh checkpoint replay; frozen-kernel controls are evaluated analytically, and the remaining experimental panels reuse saved predictions. No intermediate evolution is inferred from endpoints.

Learning in physical time. Figure 1 follows the paired-label circle task with two hidden tanh layers of width 2048. Dense training and orders $q = 1, 3, 7$ have predictions saved at the same 39 physical times between 0 and 80, with denser, approximately logarithmic spacing from 0.5 onward. Four snapshots show the function developing from the common initialization. Over all shared snapshots, on 2048 uniform circle queries against the finer dense replay, the RMS discrepancy stays below 0.071 for $q = 1$, 0.0059 for $q = 3$ and 0.00018 for $q = 7$; the full curves are in Figure 9 (Appendix I), together with the numerical sensitivity scale obtained by summing the dense and closure coarse/fine prediction changes, which the smallest discrepancies approach. The replay

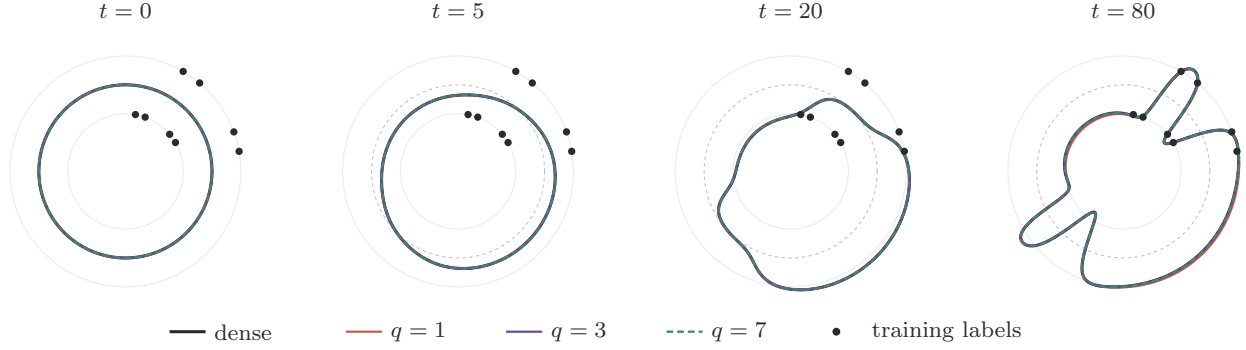


Figure 1: Learning at common physical times. Paired-label task, two hidden tanh layers, width 2048: saved functions of dense training and of response memory at orders $q = 1, 3, 7$ at the same physical times $t = 0, 5, 20, 80$. Radius is $3 + f(\theta)$, the dashed ring marks $f = 0$ and the light rings $f = \pm 1$; black dots are training labels. The RMS discrepancies at all 39 shared times are in Figure 9.

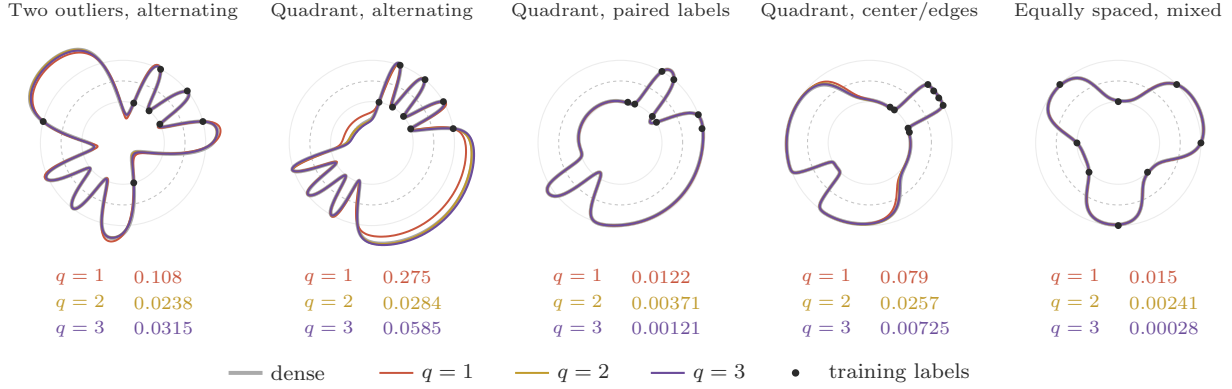


Figure 2: The fitted function across five tasks. Three hidden tanh layers, width 4096, and memory orders $q = 1, 2, 3$, at each model’s training-MSE 10^{-3} endpoint. Angle represents input direction; radius is $3 + f(\theta)$, the dashed ring marks $f = 0$ and the light rings $f = \pm 1$. The thick grey curve is dense training, coloured curves are response memory, and black dots are training labels. The numbers under each panel are RMS differences from dense on 8192 uniform circle points.

preserves the original initialization and uses float64 adaptive Heun integration at two tolerances for every model. These discrete observations illustrate tracking; they do not certify the error between observations.

Fitted functions across tasks. With eight labelled points on a circle, many functions fit the labels, so agreement at training points says little. We compare the selected function on 8192 angles for five tasks with alternating labels, clustered points and large unlabelled arcs. Figure 2 displays the three-hidden-layer results; the two-hidden-layer gallery is in Appendix H. The same initialization is used within each task and architecture. A supplementary four-hidden-layer ReLU example on the sphere appears in Appendix G.

In two hidden layers (Figure 6, Table 2 in Appendix H), the fitted-function discrepancy falls by 1.5–3 orders of magnitude from $q = 1$ to $q = 7$ on every task. In three hidden layers, $q = 2$ and $q = 3$ have RMS differences below 0.06 on all five tasks, and below 0.01 on three. Order is not monotone at these small values: $q = 2$ beats $q = 3$ on both alternating tasks; the observed numerical

sensitivity is two orders of magnitude below these differences, and no power law is fitted (Figure 10, Appendix I). The hidden features move substantially: the *absolute* activation RMS changes from initialization are 0.35–0.65, 0.38–0.74 and 0.46–0.68 in the three layers, across the 20 finest dense and closure runs. The moving state at $q = 3$ is about 3 MiB against 256 MiB for the dense learned matrices; retained initialized matrices are additional.

Frozen dictionaries built at initialization are a different alternative: on the outlier task they reach RMS 0.55–1.38 with about 45 vectors, against 0.073 for 48 response moments. Together with the factor control that follows, this motivates both parts of the method: evolving response directions and an evolution law derived from the network’s own history.

Capturing the learned function requires the right dynamics. Figure 3 compares dense training with its full initialization-frozen empirical tangent kernel, directly trained factors, and response memory. All four fit the training labels to MSE approximately 0.001, yet select different functions between those labels. On the alternating-label quadrant task, the frozen kernel has whole-circle RMS discrepancy 171.03 from dense, the two factor seeds have discrepancies 0.6766 and 1.5413, and memory at $q = 3$ has discrepancy 0.04561. The kernel includes every parameter block with the canonical mobilities; its construction and all five task comparisons are given in Appendix J and figure 14.

Both response memory and directly trained factors retain the same $W_0^{(2)}$ and have evolving low-rank learned corrections. Their dynamics differ: ordinary Euclidean factor gradient flow does not induce canonical matrix gradient flow. Figure 12 summarizes the comparison across tasks, and Figure 13 shows the fitted functions on the paired-label task at correction rank bound 24. Memory has whole-circle RMS 0.00282 from dense, against 0.375 and 0.204 for the two factor seeds. Across all five tasks and three ranks, memory has smaller discrepancy in all 29 resolved fitted comparisons; one capped factor run is excluded. Twenty-eight wins exceed a factor of two. Together, the controls show that the reproduced predictor goes beyond the initial tangent model and this rank-matched factor flow. They support the response-memory evolution law, without asserting a universal comparison with kernels or low-rank algorithms.

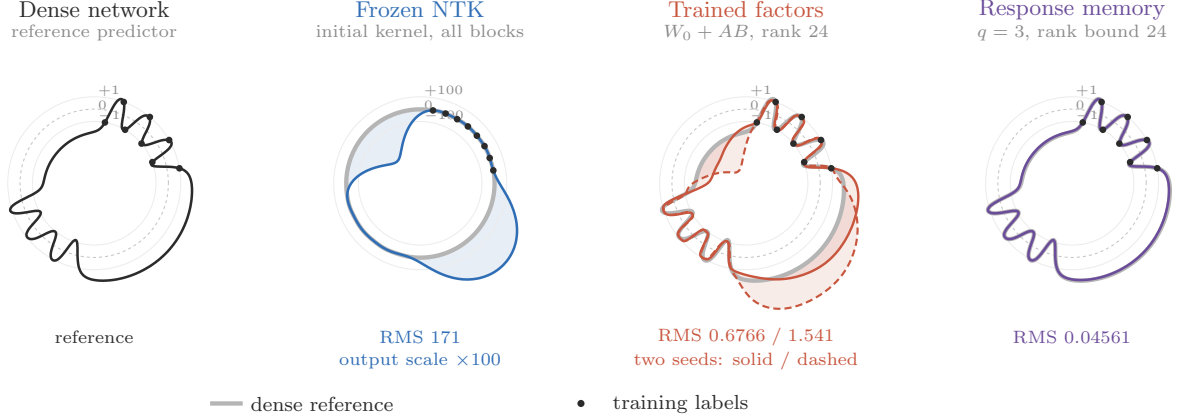
MNIST. With 100 training images and width 4096, held-out discrepancies are 0.00325, 0.00108 and 0.00120 for $q = 1, 2, 3$, retaining the small deterioration from $q = 2$ to $q = 3$; the predictions and their signed residuals are shown in Figure 11 (Appendix I). All models have identical test accuracy (94.7%). With 1000 training images and width 1024, the discrepancies are 0.019, 0.0019 and 0.0012, with 97.5% accuracy for dense, $q = 2$ and $q = 3$. Prediction fidelity and label accuracy are separate observables.

Two clocks. The joint clock’s better rate is asymptotic. At $q = 5$ it improved some cases (tanh alternating quadrant $0.0066 \rightarrow 0.0035$; GELU outliers $0.0020 \rightarrow 0.0010$) and was much worse on another (GELU alternating quadrant $0.013 \rightarrow 1.00$, reduced to 0.10 by halving the step). Over an 82-configuration sweep, on the 65 configurations where dense and all orders fit, the joint clock’s median discrepancies were 0.031, 0.013 and 0.0066 for $q = 1, 2, 3$.

5 Related work and comparison

We organize prior work by where it sits on the axis of the introduction: whether it targets the specified deep nonlinear feature-learning flow and, when it does, how its evolving state is represented. Table 1 places one representation from each description on that axis: what it stores and which

Quadrant, alternating



Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time. RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 3: Fitting the labels does not determine the learned function. Alternating-label quadrant task, eight training inputs, two hidden tanh layers and width 2048. From left: dense training, the full frozen empirical NTK, directly trained factors at rank 24 (two seeds), and response memory at $q = 3$ with rank bound 24. All start from the same physical network and are evaluated at their own training-MSE 0.001 endpoints. Gray curves show the dense reference in each panel; black dots are training labels. Angle is input direction and radius is linear in output. The NTK panel uses a 100-times wider output scale, including its reference curve and labels; signed ticks show original output units. RMS values are unscaled discrepancies from dense over 8192 circle queries, not errors against unseen labels. The complete five-task comparison appears in Figure 14.

accuracy certificate addresses the specified target. Section 3.6 gives our all-time trade-off; Appendix F details the additional hypotheses needed to compare numerical solvers at a common tolerance.

Off the axis: descriptions that give up one property. Exact solutions of deep linear networks keep depth and feature learning but not nonlinearity (Saxe et al., 2014); their infinite-width limit is a gradient flow on ℓ^2 operators added to the frozen initialization (Chizat et al., 2024), the cleanest example of an operator description. Wasserstein and mean-field limits describe one-hidden-layer networks as the gradient flow of a measure over neurons (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); they keep nonlinear feature learning but not depth. The neural tangent kernel and lazy training keep depth and nonlinearity but freeze the features, reducing training to a fixed kernel (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple because of what it sacrifices, and each is exact about a different object. These regime changes can leave a discrepancy that increasing width does not remove. This concerns the learning dynamics, not whether a shallow or kernel model can represent a particular fitted function.

On the axis, paying in history. Tensor Programs IV proves feature-learning limits at each fixed number of steps, with a recursion that includes variables indexed by every earlier step (Yang and Hu, 2021). Feature-learning DMFT evolves two-time activation, gradient and response kernels, with memory quadratic and time cubic in the number of time points (Bordelon and Pehlevan, 2022), and its finite-width extension adds perturbative fluctuations (Bordelon and Pehlevan, 2023). Direct two-time storage is smaller than dense $O(n^2)$ weight storage per layer when the number of samples times the number of time steps is well below the width. Runtime additionally depends on sampled

Table 1: Where descriptions retain the evolving state. Counts are per hidden layer, omitting fixed sources and common outer weights; N is the number of retained training times. Error claims must refer to the same deep feature-learning target and horizon before their exponents can be compared. The last column records the available scope rather than transferring a theorem from a different scaling.

Description	Moving state	Accuracy relative to the specified flow
Deep linear; shallow mean field; frozen NTK	Model dependent	Change at least one of depth, nonlinearity or feature learning.
Canonical dense network	n^2	All-time convergence as $n \rightarrow \infty$ in our small-label regime; no numerical width rate here.
Stored update histories	mnN	Exact representation of the chosen discrete updates; continuous-time error requires a numerical estimate.
Tensor Programs	Representation dependent	Fixed-program population laws; a solver and its numerical error must be specified.
Two-time DMFT solver	$m^2 N^2$ kernels	Path sampling, time discretization and self-consistency errors require separate control.
Tangent hierarchy through q	$\sum_{j < q} m^j$	Published width/horizon estimate is in a different scaling; a matched hierarchy requires its own tail and stability bound.
Response memory, speed clock	mnq	All-time $\omega(q) + o_{\text{Pr}}(1)$ in Theorem 3.1; $\omega(q) = q^{-2+o(1)}$.
Response memory, joint clock	$mnq + q^2$	$O_{n,T}(q^{-2})$ on compact horizons (Theorem D.1).

single-site paths and self-consistency sweeps (Appendix F.3). Multilayer mean-field theory and Neural Feature Flow evolve fields on neuronal ensembles (Nguyen and Pham, 2023; Fang et al., 2020). These keep all three properties within their respective parameterizations. They compress width: DMFT, for instance, has kernel state independent of the number of neurons. Direct history solvers retain time-indexed state; field descriptions instead retain functions on population spaces. Their representation and approximation costs must be specified separately. Our result is complementary: it keeps the neurons and compresses time.

On the axis, paying in tuples of samples. The Neural Tangent Hierarchy (Huang and Yau, 2020) evolves the outputs together with the tangent kernels of order $2, \dots, q$, the j -th a function of j training points, and closes by freezing the top kernel. The hierarchy is an exact identity in any parametrization, and above order two it captures feature learning; the neural tangent kernel is its lowest truncation. What it pays is combinatorial in the samples: the moving state through order q is $\sum_{j=1}^{q-1} m^j \sim m^{q-1}$ scalars beneath a frozen tensor of m^q entries (a direct count rather than a statement of that work), against a state linear in m here. Its published guarantee uses width-suppressed higher kernels: for even q it gives an $n^{-q/2}$ factor multiplied by time-dependent terms, on a width-dependent horizon and under specified data and regularity assumptions. Those estimates cannot simply be transferred to the present scaling, where feature and kernel evolution can remain order one. The matched hierarchy still captures feature learning, but its truncation error must be controlled through higher-kernel growth (Appendix F.2). Response memory instead controls the learned parameter trajectory and whole-input prediction in Theorem 3.1, with state linear in m and q . NTH’s higher orders do capture feature learning; the issue in a quantitative comparison is their tail in the same parameterization.

Memory and low rank. HiPPO provides online Legendre projection of a given signal (Gu et al., 2020); we apply it to endogenous, coupled training histories and prove stability of the resulting feedback loop. Low-rank training methods constrain the learned increment to evolving factors (Zhao et al., 2023); response-memory factors are causal summaries of the actual forward and backward histories, and a directly trained factorization of matched rank performs far worse (Section 4).

The comparison is therefore between representations and their certificates. The proved improvement here is the all-time compression in Corollary 3.2. A numerical ranking by powers of an accuracy tolerance additionally needs width, discretization and stability rates for each implementation; Appendix F states those obligations explicitly.

6 Discussion: toward an idealized model of learning

We asked whether the limiting process of deep feature learning can be approximated by a smaller evolving state. The central result gives a constructive answer in a canonical small-label regime. Each neuron remembers finitely many paired response moments, and these memories evolve with the network they reconstruct. The resulting approximation follows the whole training trajectory and its selected test function. Any diverging sublinear order reaches the dense population predictor with a vanishing fraction of the dense learned state.

The loss-decay result is part of this mechanism. Fitting is established for every order before comparison with the dense model: it bounds the total learning activity, keeps the readout-feature Gram nondegenerate, and makes all-time error control possible. The role of small labels can also be expressed as a bound on feature-learning strength through the exact rescaling in Corollary 3.3. This clarifies the regime without replacing the canonical theorem by a different training algorithm.

The next questions are quantitative width concentration, a unique fixed-order population moment law, and comparable all-time bounds when feature movement is no longer small. The current theorem requires zero readout, a positive initial feature-Gram gap and smooth gates; exact ReLU is outside its activation class. Sample dependence remains linear, so a continuum of training inputs needs a further representation. Finally, the initialized random operator remains a fixed dense environment: compressing that environment while preserving its forward and adjoint reuse is a different problem. These questions concern the remaining parts of the learning state; the response history itself already admits an autonomous, quantitatively controlled compression.

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A Proof of the all-time theorem

We prove Theorem 3.1 for layer-dependent activations. Write $g_\ell = (\phi_\ell)'$, and fix bounds

$$|\phi_\ell(0)| \leq a, \quad |g_\ell| \leq s, \quad |g_\ell(u) - g_\ell(v)| \leq j|u - v|.$$

All constants below may depend on these bounds, the fixed data and depth, and the initial Gram margin. They do not depend on n, q , or time. Constants denoted by C may increase from line to line. For sample vectors write $\|v\|_m^2 = m^{-1} \sum_a v_a^2$, so $Y = \|y\|_m$. Vector and matrix norms at finite

width are ordinary Euclidean and Frobenius norms; every normalization by n is displayed. The distance d_n is the sum of scaled block norms in (3). We first prove uniform fitting at every order, then construct the dense Gaussian reference and transfer its carrier tails, and finally use these tails to sharpen the closure error.

A.1 Initialization and polynomial projection

Let $v_a = x_a/\sqrt{d}$, $X = \max_a \|v_a\|_2$, and define the deterministic uncentered feature covariances

$$\begin{aligned} Q_{ab}^{(1)} &= \mathbb{E}[\phi_1(G^\top v_a)\phi_1(G^\top v_b)], & G &\sim N(0, I_d), \\ Q_{ab}^{(\ell)} &= \mathbb{E}[\phi_\ell(Z_a)\phi_\ell(Z_b)], & Z &\sim N(0, Q^{(\ell-1)}), \quad \ell \geq 2. \end{aligned} \quad (19)$$

Choose $\lambda > 0$ with $Q^{(L)}/m \succeq 2\lambda I_m$. There are fixed K_0, A_0, C_0 such that the events

$$\mathcal{G}_n = \left\{ \begin{array}{l} \max_{\ell \geq 2} \|W_0^{(\ell)}\|_{\text{op}} \leq K_0, \quad \max_a \frac{\|z_a^{(1)}(0)\|_2}{\sqrt{n}} \leq A_0, \\ \frac{\|W_0^{(1)}\|_F}{\sqrt{n}} \leq C_0, \quad \Gamma_w(0) := \frac{H_L(0)^\top H_L(0)}{mn} \succeq \lambda I_m \end{array} \right\} \quad (20)$$

have probability tending to one. Here H_L has feature columns indexed by the training samples. Indeed, a $1/4$ -net of the unit sphere has at most 9^n points, and approximation of both vectors in a bilinear form gives

$$\Pr\{\|W_0^{(\ell)}\|_{\text{op}} > K_0\} \leq 2 \cdot 9^{2n} e^{-nK_0^2/8} \rightarrow 0$$

for a sufficiently large fixed K_0 . The first-layer bounds follow from the law of large numbers at fixed d, m .

For completeness, $H_\ell(0)^\top H_\ell(0)/n \rightarrow Q^{(\ell)}$ in probability. At the first layer this is a law of large numbers. Conditionally on all previous layers, the rows of the next preactivation matrix are independent $N(0, H_{\ell-1}^\top H_{\ell-1}/n)$ vectors. Linear growth, $|\phi(u)| \leq a + s|u|$, bounds each activation's Gaussian fourth moment by $8a^4 + 24s^4 Q_{aa}^2$. Conditional Chebyshev therefore bounds each covariance-entry error by $C(M)/(n\epsilon^2)$ whenever the previous covariance has operator norm at most M . Finally the map $Q \mapsto \mathbb{E}[\phi(Q^{1/2}G)_a \phi(Q^{1/2}G)_b]$ is continuous, including at singular Q : positive square roots converge, and a constant multiple of $1 + \|G\|_2^2$ is an integrable majorant on bounded covariance sets. Induction proves the assertion and hence the Gram part of (20).

The input criterion stated with the theorem also follows directly. If the nonzero v_a are pairwise nonproportional and $\sum_a c_a \phi_1(v_a^\top u) = 0$ for all u , fix a with $c_a \neq 0$. For each $b \neq a$, choose $u_b \perp v_b$ with $u_b \cdot v_a \neq 0$. Applying all $m-1$ finite differences in the directions $t_b u_b$ eliminates every term but the a th, giving

$$\Delta_{s_1} \cdots \Delta_{s_{m-1}} \phi_1(t) = 0 \quad \text{for all } t, s_1, \dots, s_{m-1}.$$

Convolve with a smooth compactly supported approximate identity and differentiate once in each s_i at zero. The mollified function is a polynomial of degree at most $m-2$. Its coefficients converge, by interpolation at $m-1$ distinct points, so the original continuous function is such a polynomial. Thus a nonpolynomial first activation makes the ridge functions independent. A null quadratic form of $Q^{(1)}$ would give a continuous ridge combination vanishing almost everywhere under a full-support Gaussian, hence everywhere, and is impossible. If $Q \succ 0$ and ψ is continuous and nonconstant, the identity $\sum_a c_a \psi(Z_a) = 0$ for $Z \sim N(0, Q)$ likewise holds everywhere; varying one coordinate shows $c_a = 0$. This propagates positivity to all layers. For $m=1$, a nonzero input and any nonconstant first activation suffice. Alternatively linearly independent inputs give a positive first preactivation

covariance, so any nonconstant activation in every layer suffices. A globally Lipschitz nonaffine function is nonpolynomial. These are sufficient criteria; the proof below only needs the actual gap in (19).

Lemma A.1 (Projection estimates used in the bootstrap). *Let Π_q^A be orthogonal projection onto polynomials of degree below q in $L^2(0, A; \mathcal{H})$, for a real Hilbert space $\mathcal{H}$. For $u \in H^1(0, A; \mathcal{H})$ and bounded measurable b ,*

$$\|(I - \Pi_q^A)u\|_{L^2}^2 \leq \frac{1}{q(q+1)} \int_0^A \xi(A - \xi) \|u'(\xi)\|^2 d\xi, \quad (21)$$

$$\|u(A) - (\Pi_q^A u)(A)\| \leq C\sqrt{A/q} \|u'\|_{L^2}, \quad \|(\Pi_q^A b)(A)\| \leq C\sqrt{q} \|b\|_{L^\infty}. \quad (22)$$

If a history u is fixed as its right endpoint grows, then

$$\frac{d}{dA} \|(I - \Pi_q^A)u\|_{L^2(0, A)}^2 = \|u(A) - (\Pi_q^A u)(A)\|^2 \quad \text{for almost every } A. \quad (23)$$

Proof. The normalized shifted Legendre modes e_k satisfy $[\xi(A - \xi)e'_k]' = k(k+1)e_k$. Integration by parts and Bessel's inequality applied to the orthonormal family $\sqrt{\xi(A - \xi)}e'_k/\sqrt{k(k+1)}$ give (21); the omitted modes have $k(k+1) \geq q(q+1)$. Smooth approximation and an orthonormal expansion in $\mathcal{H}$ give the stated generality.

On $[0, 1]$ the endpoint kernel is $K_q(u) = \sum_{k < q} (2k+1)p_k(u) = (p'_q(u) + p'_{q-1}(u))/2$. Integration by parts yields

$$u(A) - (\Pi_q^A u)(A) = \frac{1}{2} \int_0^A [p_q(\xi/A) + p_{q-1}(\xi/A)] u'(\xi) d\xi.$$

The squared L^2 norm of this multiplier is $A[(2q+1)^{-1} + (2q-1)^{-1}]/4$, proving the first endpoint bound. For the other one it suffices to check $\int_0^1 |K_q| \leq C\sqrt{q}$. Here is a short verification of the needed Legendre variation estimate. For the ordinary Legendre polynomial L_k , set $v(\theta) = \sqrt{\sin \theta} L_k(\cos \theta)$ and $Q(\theta) = (k+1/2)^2 + (4 \sin^2 \theta)^{-1}$. The Legendre equation gives $v'' + Qv = 0$ and $(v^2 + (v')^2/Q)' = -Q'(v')^2/Q^2 \geq 0$ on $(0, \pi/2)$. The values $L_{2r}(0) = (-1)^r \binom{2r}{r}/4^r$, $L_{2r+1}(0) = 0$, and $L'_k(0) = kL_{k-1}(0)$ show that this energy at $\pi/2$ is at most C/k , using $\binom{2r}{r}/4^r \leq (r+1)^{-1/2}$. Consequently, on $1/k \leq \theta \leq \pi/2$,

$$|\partial_\theta L_k(\cos \theta)| \leq Ck^{-1/2} \{k(\sin \theta)^{-1/2} + (\sin \theta)^{-3/2}\}.$$

Its integral is at most $C\sqrt{k}$. On $[0, 1/k]$, use $|L'_k| \leq k(k+1)/2$; this follows by summing $L'_{r+1} - L'_{r-1} = (2r+1)L_r$ and $|L_r| \leq 1$ on $[-1, 1]$. The last bound follows, for example, by integrating $(\cos \theta + i \sin \theta \cos \alpha)^r$ over $0 \leq \alpha \leq \pi$. Parity handles the other half. Thus $\text{Var}(L_k) \leq C\sqrt{k}$, which proves the kernel bound; $q = 1$ is immediate.

Finally differentiate the minimum least-squares error on $[0, A]$. The endpoint contributes the right side of (23). The other term pairs the projection residual with the derivative of a polynomial of degree below q and vanishes by orthogonality. Moment coordinates and the invertible polynomial Gram matrix justify absolute continuity of its coefficients on compact endpoint intervals. This argument only needs square-integrable histories for the last identity. $\square$

A.2 Fitting and continuation for every memory order

Fix an initialization in $\mathcal{G}_n$ and put $D_0 = K_0 + 1$. On the tube $\|W^{(\ell)}\|_{\text{op}} < D_0$ for $\ell \geq 2$ and $\max_a \|z_a^{(1)}\|_2/\sqrt{n} < A_0 + 1$, define

$$M_1 = a + s(A_0 + 1), \quad M_\ell = a + sD_0 M_{\ell-1}.$$

Linear growth gives $\|h_a^{(\ell)}\|_2/\sqrt{n} \leq M_\ell$. Stop the local closure at first tube exit or activity $S = 4Y/\lambda$, reducing the eventual Y_* so $S \leq 1$. Since $w(0) = 0$, the readout equation and backward recursion give

$$\frac{\|w\|_2}{\sqrt{n}} \leq 2M_L S, \quad \max_a \frac{\|\delta_a^{(\ell)}\|_2}{\sqrt{n}} \leq \beta_\ell, \quad \beta_L = 2sM_L S, \quad \beta_\ell = sD_0\beta_{\ell+1}. \quad (24)$$

Thus every $\beta_\ell \leq CY$. In this subsection responses and residuals belong to the closure, with hats suppressed. Write $c_a = r_a/\rho$, $b_a^{(\ell)} = c_a\delta_a^{(\ell)}$, and $A = \tau(t) \leq 1 + S \leq 2$. The forward prefix is constant and the backward prefix is zero. Projection contraction in the reconstruction gives

$$\|\widehat{W}^{(\ell)} - W_0^{(\ell)}\|_F \leq 2M_{\ell-1}\beta_\ell\sqrt{(1+S)S} = O(Y^{3/2}), \quad (25)$$

$$\frac{\|\widehat{W}^{(1)} - W_0^{(1)}\|_F}{\sqrt{n}} \leq 2X\beta_1 S = O(Y^2). \quad (26)$$

For the first inequality, the backward history's averaged squared norm is at most $S\beta_\ell^2$, the forward one at most $(1+S)M_{\ell-1}^2$, and $\|uv^\top/n\|_F = (\|u\|_2/\sqrt{n})(\|v\|_2/\sqrt{n})$. These bounds keep both tube boundaries strictly away for small Y .

Differentiating the reconstruction gives exactly

$$\dot{\hat{\theta}} = F(\hat{\theta}) + E, \quad E_1 = E_w = 0, \quad E_\ell = \frac{2\rho}{mn} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top, \quad (27)$$

where stars denote projected endpoint values. By (23), zero initial projection errors, and Cauchy-Schwarz, its accumulated absolute norm satisfies

$$\int_0^t \|E_\ell(u)\|_F du \leq \frac{2}{mn} \sum_a \|(I - \Pi_q^A)b_a^{(\ell)}\|_{L^2(0,A)} \|(I - \Pi_q^A)h_a^{(\ell-1)}\|_{L^2(0,A)}. \quad (28)$$

This bounds total absolute velocity error, not just the error in a signed history integral.

Set

$$Z_\ell(t) = \frac{1}{mn} \sum_a \int_0^{\tau(t)} \|\partial_\xi h_a^{(\ell)}\|_2^2 d\xi.$$

First $Z_1 \leq 4Ss^2X^4\beta_1^2$. Endpoint evaluation on degree-below- q polynomials has norm $q/\sqrt{A}$ because $\sum_{k < q} (2k+1) = q^2$. The averaged backward endpoint error is consequently at most $(q+1)\beta_\ell$ in normalized Euclidean norm. Equations (21)–(27) imply

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 du \leq 2(1+S)^2\beta_\ell^2 Z_{\ell-1}(t).$$

The clock chain rule is

$$\partial_\xi h_a^{(\ell)} = g_\ell(z_a^{(\ell)}) \odot \{(F_\ell/\rho + E_\ell/\rho)h_a^{(\ell-1)} + \widehat{W}^{(\ell)}\partial_\xi h_a^{(\ell-1)}\}.$$

As $\|F_\ell/\rho\|_F \leq 2\beta_\ell M_{\ell-1}$, the preceding estimate proves inductively

$$\begin{aligned} Z_1 &\leq Z_1^* := 4Ss^2X^4\beta_1^2, \\ Z_\ell &\leq Z_\ell^* := 3s^2 [4S\beta_\ell^2 M_{\ell-1}^4 + \{D_0^2 + 2(1+S)^2\beta_\ell^2 M_{\ell-1}^2\}Z_{\ell-1}^*] \leq CY^3. \end{aligned} \quad (29)$$

The factors $M_{\ell-1}^4$ and $M_{\ell-1}^2$ retain the contribution of unbounded activation values. No feature coordinate maximum is used. The sharper endpoint bounds (22) now give forward error at most $CY^{3/2}/\sqrt{q}$ and backward error at most $CY\sqrt{q}$. Their product in (27) yields

$$e_E(t) := \sum_{\ell=2}^L \|E_\ell(t)\|_F \leq CY^{5/2}\rho(t), \quad \int_0^t e_E du \leq CY^3/q. \quad (30)$$

The second estimate also follows from (28), the $CY^{3/2}/q$ forward tail and the $CY^{3/2}$ backward history norm.

The tangent Gram in the residual equation is

$$\Gamma_{ab} = \frac{1}{m} \left[\frac{(h_a^{(L)})^\top h_b^{(L)}}{n} + \frac{(\delta_a^{(1)})^\top \delta_b^{(1)}}{n} v_a^\top v_b + \sum_{\ell=2}^L \frac{(\delta_a^{(\ell)})^\top \delta_b^{(\ell)}}{n} \frac{(h_a^{(\ell-1)})^\top h_b^{(\ell-1)}}{n} \right]. \quad (31)$$

Every summand is a parameter-gradient Gram, so $\Gamma \succeq \Gamma_w$. Equation (29) gives

$$\frac{\|H_L(t) - H_L(0)\|_F}{\sqrt{mn}} \leq \sqrt{SZ_L^*}, \quad \|\Gamma_w(t) - \Gamma_w(0)\|_{\text{op}} \leq 2M_L \sqrt{SZ_L^*} \leq CY^2.$$

After reducing Y_* , $\Gamma \succeq \lambda I/2$ and $\|\Gamma\|_{\text{op}} \leq G_*$ on the stopped interval. For the ordinary prediction Jacobian J , $(JE)_a = \sum_{\ell \geq 2} (\delta_a^{(\ell)})^\top E_\ell h_a^{(\ell-1)}/n$, whence

$$\dot{r} = -2\Gamma r + JE, \quad \|JE\|_m \leq CY e_E \leq CY^{7/2}\rho.$$

Choose Y_* so $CY_*^{7/2} \leq \lambda/2$ and put $\kappa = \lambda/2$. Taking the inner product with r/ρ proves

$$Y e^{-\Lambda t} \leq \rho(t) \leq Y e^{-\kappa t}, \quad \int_0^t \rho du \leq Y/\kappa = S/2, \quad (32)$$

for a fixed finite Λ . Thus the activity boundary is also excluded.

These arguments legitimately use the clock before its lower residual bound has been proved. The raw moment vector field is locally Lipschitz for $\tau > 0$, and all its velocities vanish at zero residual. Local uniqueness backwards from such an equilibrium excludes a first finite zero on a nonstationary regular solution. At fixed n, q , the physical bounds above and the integral moment formulas bound the entire raw state in a compact subset of $\tau > 0$. A finite maximal endpoint would therefore extend. This proves global existence, uniqueness and (32) for every $q \geq 1$. The dense flow has the same proof with $E = 0$ and hidden displacement at most $2S\beta_\ell M_{\ell-1}$. All reductions of Y_* impose finitely many inequalities whose left sides vanish with Y ; none involves width or order.

The update equations, (30), and forward chain rule also give, for either path with its own residual,

$$\begin{aligned} \frac{\|\dot{W}^{(1)}\|_F}{\sqrt{n}} + \sum_{\ell=2}^L \|\dot{W}^{(\ell)}\|_F &\leq CY\rho, & \frac{\|\dot{w}\|_2}{\sqrt{n}} &\leq C\rho, \\ \max_{\ell,a} \frac{\|\dot{z}_a^{(\ell)}\|_2 + \|\dot{h}_a^{(\ell)}\|_2}{\sqrt{n}} &\leq CY\rho, & \int_t^\infty \rho(u) du &\leq \rho(t)/\kappa. \end{aligned} \quad (33)$$

Consequently physical parameters converge to finite interpolating limits. The residual equation further gives $\|\dot{r}\|_m \leq C\rho$ and $\|\dot{c}\|_m \leq C$. When $Y = 0$, both paths are stationary and all conclusions are immediate.

A.3 The Gaussian reference and its finite-width transfer

We give the Gaussian construction needed below, including the fixed computation input. Its purpose is to control full dense backward carriers, including the trained readout. All Gaussian bounds in this subsection initially concern population reference computations; no Gaussian law for the coordinates of a trained finite network is assumed.

Lemma A.2 (Fixed Gaussian computations). *Consider a fixed finite list of typed vector instructions using independent matrices with entries $N(0, 1/n)$, both orientations of each same matrix, independent identically distributed root tuples in each layer, deterministic linear combinations, and C^1 coordinate maps with bounded first derivatives. Root tuples in different layers are independent and have finite second moments. Every same-layer tuple has a deterministic limiting empirical law, in probability in Wasserstein distance of order two. Finite collections of such programs sharing the arrays converge jointly.*

The limiting forward and reverse actions of one initialized matrix are

$$\mathcal{W}h = \xi_h + \sum_{s: \mathcal{W}^*u_s \text{ earlier}} u_s \mathbb{E}[\partial_{\xi_s} h], \quad \mathcal{W}^*u = \zeta_u + \sum_{r: \mathcal{W}v_r \text{ earlier}} v_r \mathbb{E}[\partial_{\xi_r} u]. \quad (34)$$

The centered Gaussian source covariances are $\mathbb{E}[\xi_h \xi_v] = \mathbb{E}[hv]$ and $\mathbb{E}[\zeta_u \zeta_v] = \mathbb{E}[uv]$. Distinct matrix/orientation source groups are independent of one another and of the roots. Each derivative is of the unrolled scalar expression in named source coordinates, holding deterministic coefficients, covariances and previously computed expectations fixed. The statement allows singular source covariances.

The value conclusion also holds for continuous coordinate maps of at most linear growth. If roots are sub-Gaussian, activations are C^2 with bounded first two derivatives, and the extra instructions are products $b(z)p$ with bounded C^1 functions b, b' , these products also obey the response rule (34).

Proof. We record the conditioning argument to specify precisely the result being used. For one matrix W , collect previous constraints as $WV = Y$ and $W^\top U = Q$. Conditional on the computation transcript, query inputs are fixed. Revealing a new answer conditions only the queried matrix's current Gaussian factor on one more linear constraint; coordinate operations reveal no additional randomness. Induction therefore leaves the residual Gaussian factors for distinct matrices conditionally independent. When the two query Grams are invertible, Gaussian orthogonal projection gives

$$W | \mathcal{H} \stackrel{d}{=} Y(V^\top V)^{-1}V^\top + U(U^\top U)^{-1}Q^\top P_{V^\perp} + P_{U^\perp} \widetilde{W} P_{V^\perp}.$$

Indeed the displayed mean solves both constraints, using $U^\top Y = Q^\top V$, and is Frobenius-orthogonal to their common homogeneous solution space. The last term is the Gaussian on that space. For a new input h , write $\alpha_n = (V^\top V)^{-1}V^\top h$ and $h_\perp = h - V\alpha_n$. Then

$$Wh = Y\alpha_n + U\beta_n + \frac{\|h_\perp\|_2}{\sqrt{n}} P_{U^\perp} g, \quad \beta_n = (U^\top U/n)^{-1}(Q^\top h_\perp/n),$$

in conditional law, with independent $g \sim N(0, I_n)$. If the limiting query Grams are positive definite, all coefficients converge by induction. The removed noise has conditional expected normalized squared norm $\text{rank}(U)/n$, and hence vanishes. For bounded Lipschitz empirical tests, the remaining independent Gaussian coordinates give conditional variance at most $4\|\psi\|_\infty^2/n$. The conditional mean is an empirical continuous function of the old tuple. Second moments converge as well: the new Gaussian cross term has variance $4\sigma^2\|m\|_2^2/n^2$ when the conditional mean is m , and the Gaussian squared-norm average has variance $2/n$. Thus weak convergence and second-moment convergence,

equivalently Wasserstein convergence of order two, propagate through a matrix call. Lipschitz coordinate instructions propagate them directly.

To identify the mean, let α be the limiting regression coefficient and $h_\perp = h - \sum_r \alpha_r v_r$. Earlier reverse answers equal ζ_s plus linear combinations of the v_r ; their pairings with $h_\perp$ are therefore $\mathbb{E}[\zeta_s h_\perp]$. Gaussian integration by parts gives

$$\mathbb{E}[\zeta h_\perp] = (\mathbb{E}[u_s u_t])_{st} \mathbb{E}[\nabla_\zeta h_\perp].$$

After inserting the earlier forward decompositions in $Y\alpha$, the terms containing $\alpha_r \mathbb{E}[\nabla_\zeta v_r]$ cancel. The remaining response is $\sum_s u_s \mathbb{E}[\partial_{\zeta_s} h]$. The Gaussian part has variance $\mathbb{E}[h^2]$ and covariance $\mathbb{E}[h v_r]$ with each previous forward source. Its fresh innovation is independent of other source groups. This proves (34); interchanging the two orientations proves the reverse formula.

To remove the rank assumption, add a fresh independent input $\epsilon\chi$ at each matrix call. At fixed $\epsilon > 0$ the squared distance of the new query from the previous query span has limiting value at least ϵ^2 : the new Gaussian squared residual tends to one, and its cross term has conditional variance $O(1/n)$ on bounded norm events. All limiting query Grams are then positive definite. On the common initialized-operator event and the event $\|\chi\|_2/\sqrt{n} \leq 2$ for the finitely many added roots, finite instruction induction gives $\max_v \|v_n^\epsilon - v_n\|_2/\sqrt{n} \leq C\epsilon$. The scalar response construction is also continuous at $\epsilon = 0$. Its covariance matrices converge, so couple their source vectors by $K_\epsilon^{1/2}G$. Positive square roots are continuous even at singular matrices: bounded subsequences of the roots have limits whose square is K , and uniqueness of the positive root identifies every limit. At a fixed finite instruction, scalar values have a uniform linear envelope and their first source derivatives are bounded when earlier coefficients lie in a compact set. Dominated convergence therefore passes values, second moments and expected source derivatives to zero noise, closing this finite induction. Taking width first and then $\epsilon \downarrow 0$ proves the singular case. This argument uses no continuity of pseudoinverses.

For a continuous map F with a linear envelope, $X_j \rightarrow X$ in L^2 implies $F(X_j) \rightarrow F(X)$ in L^2 : compact uniform continuity gives convergence in probability, and the common linear envelope supplies uniform integrability. Approximate F by smooth compactly supported maps with a common linear envelope. Choose each current approximation first, then the approximation tolerance for its finite input prefix smaller than its error budget divided by that smooth map's Lipschitz constant. Matrix actions multiply same-array RMS errors by at most K_0 . Finite induction therefore removes these approximations in both the scalar and empirical laws; no uniform cutoff Lipschitz constant is needed.

For the final response assertion, replace $b(z)p$ by $b(z)\chi_R(p)$, where χ_R is a smooth clip with $|\chi_R(p)| \leq |p|$, $|\chi_R'| \leq 1$, converging locally in C^1 to the identity. The clipped coordinate maps have bounded derivatives. In a fixed scalar program the values have a linear envelope in the finite root/source list; their first source derivatives have a polynomial envelope, since each differentiated product adds at most one carrier factor. Sub-Gaussian roots and finite Gaussian source lists have all moments. Covariance-square-root coupling and uniform integrability thus pass the expected derivatives through the clips, one instruction at a time. The value limit agrees with the preceding continuous-map extension. The number of instructions is fixed throughout. $\square$

The same-array operator inequality used in this proof constructs actual population operators, rather than independent reverse maps. Include a countable family of all required finite programs, their finite unions, rational linear combinations, smooth coordinate approximations, and forward and reverse queries. Their scalar laws are consistent, since deleting unused instructions changes no finite-array node. Realize their countable joint laws, separately at each layer, and complete the generated span in L^2 . The high-probability inequality $\|W_0 v\|_2/\sqrt{n} \leq K_0 \|v\|_2/\sqrt{n}$ passes to each

finite generated span by second-moment convergence, then to its completion. Including a countable dense family of bounded smooth functions of finite tuples makes this completion the L^2 space of the generated sigma field. The finite identity $n^{-1}u^\top W_0 v = n^{-1}(W_0^\top u)^\top v$ passes to these spans and then by continuity, so the reverse action is exactly W_0^* . Increments of a hidden operator will be Hilbert–Schmidt: $(u \otimes v)h = u\mathbb{E}[vh]$ has $\|u \otimes v\|_{\text{HS}} = \|u\|_2\|v\|_2$, the counterpart of the finite Frobenius identity.

Activity-stopped population programs. Initially assume the activations are C^2 with $\|\phi'_\ell\|_\infty \leq s$ and $\|\phi''_\ell\|_\infty \leq j$. Use upper-case Z, H, P for population scalar fields, with $P_a^{(L)} = w$, $P_a^{(\ell)} = (W^{(\ell+1)})^* \delta_a^{(\ell+1)}$, and $\delta_a^{(\ell)} = g_\ell(Z_a^{(\ell)})P_a^{(\ell)}$. For an Euler program with physical steps h_k , let

$$\rho_k = \|r_k\|_m, \quad \alpha_k = h_k \rho_k, \quad u_{a,k} = r_{a,k} / \rho_k.$$

Freeze zero-residual nodes. Otherwise $|u_{a,k}| \leq \sqrt{m}$. The update formulas in activity are

$$\begin{aligned} \Delta w_k &= -\frac{2\alpha_k}{m} \sum_a u_{a,k} H_{a,k}^{(L)}, \\ \Delta W_k^{(1)} &= -\frac{2\alpha_k}{m} \sum_a u_{a,k} \delta_{a,k}^{(1)} v_a^\top, \\ \Delta W_k^{(\ell)} &= -\frac{2\alpha_k}{m} \sum_a u_{a,k} \delta_{a,k}^{(\ell)} \otimes H_{a,k}^{(\ell-1)}. \end{aligned} \tag{35}$$

At any fixed population program all coefficients here are deterministic, although calculated causally from earlier population expectations. Thus Lemma A.2 applies with these coefficients frozen in every source derivative.

Stop at activity $\sum_{k < j} \alpha_k \leq S \leq 1$ and at the physical tube used above. The same linear-growth induction gives $\|H_a^{(\ell)}\|_2 \leq M_\ell$, $\|w\|_2 \leq 2M_L S$, and $\|\delta_a^{(\ell)}\|_2 \leq C_\ell S$. Summing the rank-one updates gives first-layer and hidden increments at most CS^2 . Subtracting successive forward recursions then gives

$$\max_{\ell, a} \|H_a^{(\ell)} - H_a^{(\ell)}(0)\|_2 \leq CS^2, \quad \|\Gamma_w - \Gamma_w(0)\|_{\text{op}} \leq CS^2.$$

These estimates also hold at an affine Euler interpolation state, regarded as one final fractional step. There is therefore a fixed $S_b > 0$ for which every stopped state has strict tube margins and

$$\Gamma \succeq \lambda I/2, \quad \|\Gamma\|_{\text{op}} \leq G, \quad \|F\|_{\text{sum}} \leq V\rho. \tag{36}$$

Here the parameter norm is the sum of the L^2 first-layer and readout norms and the Hilbert–Schmidt hidden-increment norms. The residual is also bounded by a fixed constant on this tube.

A response bound uniform in the number of steps. For each hidden initialized matrix, introduce its forward sources $\xi_{a,k}^{(\ell)}$ and reverse sources $\zeta_{a,k}^{(\ell)}$. Their within-group covariances are respectively $\mathbb{E}[H_{a,k}^{(\ell-1)} H_{b,r}^{(\ell-1)}]$ and $\mathbb{E}[\delta_{a,k}^{(\ell)} \delta_{b,r}^{(\ell)}]$. Expanding the learned operators in (35) and using

(34) gives

$$\begin{aligned}
Z_{a,k}^{(\ell)} &= \xi_{a,k}^{(\ell)} + \sum_{b,r < k} F_{ak,br}^{(\ell)} \delta_{b,r}^{(\ell)}, & P_{a,k}^{(\ell-1)} &= \zeta_{a,k}^{(\ell)} + \sum_{b,r \leq k} D_{ak,br}^{(\ell)} H_{b,r}^{(\ell-1)}, \\
F_{ak,br}^{(\ell)} &= C_{ak,br}^{(\ell)} - \frac{2\alpha_r u_{b,r}}{m} \mathbb{E}[H_{b,r}^{(\ell-1)} H_{a,k}^{(\ell-1)}], & C_{ak,br}^{(\ell)} &= \mathbb{E} \frac{\partial H_{a,k}^{(\ell-1)}}{\partial \zeta_{b,r}^{(\ell)}}, \\
D_{ak,br}^{(\ell)} &= A_{ak,br}^{(\ell)} - \mathbf{1}_{r < k} \frac{2\alpha_r u_{b,r}}{m} \mathbb{E}[\delta_{b,r}^{(\ell)} \delta_{a,k}^{(\ell)}], & A_{ak,br}^{(\ell)} &= \mathbb{E} \frac{\partial \delta_{a,k}^{(\ell)}}{\partial \xi_{b,r}^{(\ell)}}.
\end{aligned} \tag{37}$$

The root equations are the first-layer and readout recursions in (35). The training-memory coefficients are bounded by $J\alpha_r/m$ for a fixed J , by the preceding RMS bounds.

For a scalar random variable let $\mathcal{N}(U) = \sup_{p \geq 2} \|U\|_p / \sqrt{p}$. If $\mathcal{N}(U) \leq B$, expansion of the exponential and $r! \geq (r/e)^r$ give $\mathbb{E} e^{U^2/(8eB^2)} \leq 4/3$, and hence $\mathbb{E} e^{t|U|} \leq (4/3)e^{2et^2B^2}$. Jensen's inequality with weights $\alpha_r/(m \sum \alpha_r)$ therefore gives, without temporal independence,

$$\mathbb{E} \exp \left(t \sum_{r < k} \frac{\alpha_r}{m} \sum_b |P_{b,r}^{(\ell)}| \right) \leq \frac{4}{3} e^{2et^2 S^2 B^2} \quad \text{if all } \mathcal{N}(P_{b,r}^{(\ell)}) \leq B. \tag{38}$$

We verify simultaneously the caps

$$\sum_{b,r \leq k} |A_{ak,br}^{(\ell)}| \leq \mathbf{a}_\ell, \quad |C_{ak,br}^{(\ell)}| \leq \mathbf{c}_\ell \alpha_r / m, \quad \mathcal{N}(H) \leq B_H, \quad \mathcal{N}(P^{(\ell)}) \leq B_{P,\ell}. \tag{39}$$

Set $f_\ell = \mathbf{c}_\ell + J$ for $\ell \geq 2$, $f_1 = 1$, $d_\ell = 1 + \mathbf{a}_{\ell+1}$ for $\ell < L$, and $d_L = 1$. Choose $K \geq 2$ depending only on the roots, source RMS bounds and activation bounds, before choosing the response caps. The field recursions imply

$$\begin{aligned}
\mathcal{N}(H_{a,k}^{(1)}) &\leq K + KS \sup_{b,r < k} \mathcal{N}(P_{b,r}^{(1)}), \\
\mathcal{N}(H_{a,k}^{(\ell)}) &\leq K + K f_\ell S \sup_{b,r < k} \mathcal{N}(P_{b,r}^{(\ell)}), \\
\mathcal{N}(P_{a,k}^{(\ell)}) &\leq K + (\mathbf{a}_{\ell+1} + JS) \sup_{b,r \leq k} \mathcal{N}(H_{b,r}^{(\ell)}), \\
\mathcal{N}(w_k) &\leq K + KS \sup_{b,r < k} \mathcal{N}(H_{b,r}^{(L)}).
\end{aligned}$$

The first two inequalities use linear growth, and $\delta = g_\ell(Z)P$ uses only bounded slope. Thus the forward fields must be bootstrapped together with the full backward carriers.

Here are the derivative estimates that close this bootstrap. Let V_k be the maximum through time k of the sum of absolute derivatives of $Z_a^{(\ell)}$ with respect to its own forward slots. Differentiating (37), the derivative row of $P_a^{(\ell)}$ is bounded by $Cd_\ell V_k$; at the top use the readout update instead. The product rule gives a row bound $C(|P_a^{(\ell)}| + d_\ell)V_k$ for $\delta_a^{(\ell)}$. Since the forward memory uses earlier times only,

$$V_k \leq 1 + C f_\ell \sum_{r < k} \alpha_r \left(d_\ell + \frac{1}{m} \sum_b |P_{b,r}^{(\ell)}| \right) V_r.$$

For a single backward source at (b, r) let D_k be the maximal absolute derivative of the lower preactivation through time k . It is zero for $k \leq r$. At the source time, differentiation of the direct

Gaussian in P gives one pulse; inserting it in the first-layer update or the next forward memory retains exactly α_r/m . All later derivatives consequently obey

$$D_k \leq C f_{\ell-1} \alpha_r / m + C f_{\ell-1} \sum_{u < k} \alpha_u \left(d_{\ell-1} + \frac{1}{m} \sum_a |P_{a,u}^{(\ell-1)}| \right) D_u.$$

Discrete Gronwall, (38), and Cauchy–Schwarz for the terminal carrier factor yield

$$\begin{aligned} \sum_{b,r \leq k} |A_{ak,br}^{(\ell)}| &\leq C(B_{P,\ell} + d_\ell) e^{C f_\ell S d_\ell + C f_\ell^2 S^2 B_{P,\ell}^2}, \\ \frac{|C_{ak,br}^{(\ell)}|}{\alpha_r / m} &\leq C f_{\ell-1} e^{C f_{\ell-1} S d_{\ell-1} + C f_{\ell-1}^2 S^2 B_{P,\ell-1}^2}. \end{aligned} \quad (40)$$

Fix one sufficiently large C in these inequalities independently of the caps. Choose, in order,

$$\mathbf{c}_2 = 4C, \quad \mathbf{c}_\ell = 4C(\mathbf{c}_{\ell-1} + J) \quad (\ell > 2), \quad \mathbf{a}_L = 4C(2K + 1), \quad \mathbf{a}_\ell = 4C(4K + 1)(1 + \mathbf{a}_{\ell+1}).$$

The $\mathbf{c}$ caps are selected from bottom to top, the $\mathbf{a}$ caps from top to bottom. Set $B_H = B_{P,L} = 2K$ and $B_{P,\ell} = 4K(1 + \mathbf{a}_{\ell+1})$ for $\ell < L$. All these constants are now fixed. Choose $S_r > 0$ so $J S_r \leq 1$, $2K S_r \leq 1$, $f_\ell S_r B_{P,\ell} \leq 1$, and every exponent in (40) is at most $\log 2$. The field recursions preserve their caps, and the response estimates improve theirs by a factor of two.

This is a chronological induction, not an assumption on future fields. At node k , the first layer and readout use only earlier nodes. Construct current forwards from bottom to top: each current $C^{(\ell)}$ uses the already constructed lower field and past lower carriers, after which $H^{(\ell)}$ uses only past same-layer carriers. Construct current backwards from top to bottom: the current upper response supplies the next carrier, and its derivative row supplies the next A . At $k = 0$ forward memory is empty and the readout is the zero root. This starts the induction. A fractional last Euler step changes only its nonnegative activity weight. We have proved, for every program stopped at $S \leq \min(S_b, S_r)$, including every affine interpolation state,

$$\max_{\ell, a} \mathbb{E} e^{c|U_a^{(\ell)}|^2} \leq C, \quad U \in \{Z, H, P, \delta\}. \quad (41)$$

The constants are independent of physical duration and number of steps.

Excluding the stop and constructing the strong flow. On the parameter tube, forward subtraction costs Cd in the parameter distance. For a reference carrier P_* , the only difficult backward product is controlled by

$$\|[g_\ell(Z) - g_\ell(Z_*)]P_*\|_2 \leq jR\|Z - Z_*\|_2 + 2s\|P_* \mathbf{1}_{\{|P_*| > R\}}\|_2. \quad (42)$$

The remaining backward difference is propagated by bounded operators and bounded gates, so new cutoff contributions add through fixed depth. The rank-one identity and (41) therefore give, for the prediction differential J from the mobility Hilbert space to sample RMS,

$$\|J(\vartheta) - J(\theta_*)\| \leq C\{(1 + R)d(\vartheta, \theta_*) + e^{-cR^2}\}. \quad (43)$$

Only the reference node needs tails. The scalar prediction is continuously differentiable in these parameter spaces. To check the possibly unbounded multiplier explicitly, its scalar Taylor remainder satisfies, for a fixed $P \in L^2$,

$$|\mathbb{E}[P\{\phi(Z + h) - \phi(Z) - g(Z)h\}]| \leq \frac{1}{2}jR\|h\|_2^2 + 2s\|P \mathbf{1}_{\{|P| > R\}}\|_2\|h\|_2 = o(\|h\|_2).$$

First send h to zero at fixed R , then R to infinity. Forward subtraction and descending this weighted scalar expansion give the displayed Jacobian; bounded-multiplier continuity proves its continuity. This does not assume that the full activation map is Fréchet differentiable from L^2 to L^2 .

An Euler segment has parameter length at most $Vh_k\rho_k$. Integrating the prediction differential on that segment and using (43) gives

$$r_{k+1} = (I - 2h_k\Gamma_k)r_k + e_k, \quad \|e_k\|_m \leq h_k\rho_k\omega(h_k), \quad \omega(h) = C \inf_{R \geq 1} \{(1+R)h + e^{-cR^2}\} \longrightarrow 0.$$

For sufficiently small maximal physical step, (36) implies

$$\rho_{k+1} \leq (1 - \lambda h_k/2)\rho_k, \quad \sum_{k < j} h_k\rho_k \leq 2Y/\lambda.$$

Choose $S = 4Y/\lambda$ and reduce Y_* so $S \leq \min(S_b, S_r)$. The first attainment of this activity cap, even by a shortened final step, contradicts the strict factor-two margin. The physical tube already has a strict margin. These deterministic Euler programs therefore extend through every prescribed physical horizon.

Put countably many refining programs on all integer horizons on the common generated spaces above. The vector-field version of (43) follows from backward subtraction and rank-one update subtraction. Comparing two Euler interpolants on $[0, T]$, with maximal steps h, h' , gives

$$\sup_{t \leq T} d(\theta^h(t), \theta^{h'}(t)) \leq C_T e^{C(1+R)T} \{(1+R)(h+h') + e^{-cR^2}\}. \quad (44)$$

The same argument holds in the Hilbert–Schmidt increment norm because $\|u \otimes v - u' \otimes v'\|_{\text{HS}} \leq \|u - u'\|_2 \|v\|_2 + \|u'\|_2 \|v - v'\|_2$. First send the meshes to zero at fixed R , then R to infinity. Since $e^{CRT-cR^2} \rightarrow 0$, completeness gives a uniform strong limit. For the passage through backward gates, if $Z_i \rightarrow Z$ in probability and $P_i \rightarrow P$ in L^2 , split $g(Z_i)P_i - g(Z)P$ into $g(Z_i)(P_i - P)$ and $[g(Z_i) - g(Z)]P$. The former converges by boundedness of g ; truncate the single integrable variable P^2 to prove convergence of the latter. Thus the vector field is continuous, and the Euler integral equations pass to the limit.

The same reference comparison against this constructed solution proves uniqueness: the discrepancy of two solutions is at most $C_T e^{CRT-cR^2}$ for every R , hence zero, up to any first tube exit. Equality to the constructed path excludes that exit. Solutions from different horizons agree on overlaps. The Gram gap survives the limit, and the exact residual equation $\dot{r} = -2\Gamma r$ gives exponential decay and finite total parameter variation, with the constants in (32)–(33) decreased or enlarged if necessary. The limit at infinite physical time is an interpolating parameter state. At each time, L^2 convergence and Fatou pass (41) to the solution; the constants are common to all times. Applying the same reasoning at the parameter limit includes $t = \infty$. The assertion bounds time marginals; it places no time supremum inside the exponential.

Finally remove classical second differentiability. Compactly supported mollification gives $\|\phi_{\ell,\epsilon} - \phi_\ell\|_\infty \leq C s \epsilon$, $\|g_{\ell,\epsilon} - g_\ell\|_\infty \leq C j \epsilon$, $\|g_{\ell,\epsilon}\|_\infty \leq s$ and $\|g'_{\ell,\epsilon}\|_\infty \leq j$. The value at zero is uniformly bounded. All response caps and activity thresholds are therefore common to sufficiently small smoothings. The initialization recursion (19) is continuous under these uniform activation perturbations, so its positive Gram margin also persists, after decreasing the common margin once. The comparison (44) between different smoothings has the additional term $C(1+R)(\epsilon + \epsilon')$ inside braces. After constructing each smoothed flow, take smoothing scales to zero at fixed R and then R to infinity. This gives the strong solution for the stated C^1 activations with Lipschitz derivatives, with the same uniqueness and Gaussian bounds.

Lemma A.3 (Dense carrier transfer). *On the events (20), let $k_{a,D}^{(L)} = w_D$ and $k_{a,D}^{(\ell)} = (W_D^{(\ell+1)})^\top \delta_{a,D}^{(\ell+1)}$ for $\ell < L$. Define*

$$H_n(M, t) = \sum_{\ell=1}^L \max_a \frac{\|k_{a,D}^{(\ell)}(t) \mathbf{1}_{\{|k_{a,D}^{(\ell)}(t)| > M\}}\|_2}{\sqrt{n}}, \quad Z_n(M) = \int_0^\infty \rho_D(t) H_n(M, t) dt,$$

and set $Z_n(M) = 0$ off $\mathcal{G}_n$. There are constants $c, C > 0$ and dense-only random variables $a_n \geq 0$, with $a_n \rightarrow 0$ in probability, such that, simultaneously for all integer $M \geq 1$,

$$Z_n(M) \leq Ce^{-cM^2} + a_n. \quad (45)$$

On each finite physical horizon, dense finite-width predictions on any fixed finite list of training or passive test inputs converge in probability, uniformly in time, to the predictions of the unique strong population flow just constructed.

Proof. Fix a finite horizon T , a reference cutoff R , and a finite population Euler mesh. Evaluate its expanded initialized-matrix calls on the actual finite Gaussian arrays, keeping every residual and scalar contraction equal to its deterministic population value. This is a fixed finite program to which Lemma A.2 applies. Reconstruct finite proxy parameters by the rank-one updates with those oracle vectors. For example a proxy action on an oracle feature differs from its prescribed action by a finite sum of terms of the form

$$-\frac{2\alpha_r u_{b,r}}{m} \delta_{b,r}^{(\ell)} \left\{ \frac{(h_{b,r}^{(\ell-1)})^\top h_{a,k}^{(\ell-1)}}{n} - \mathbb{E}[H_{b,r}^{(\ell-1)} H_{a,k}^{(\ell-1)}] \right\}.$$

Each brace converges to zero; all involved vector RMS norms are bounded in probability. Forward recomputation is therefore consistent. For backward recomputation use (42) against the oracle carriers. Continuous quadratic-growth cutoff measurements of their joint empirical laws converge by the lemma. At fixed cutoff the consistency errors vanish, and the limiting cutoff tails are Ce^{-cR^2} by (41). Sending that cutoff to infinity proves RMS consistency of the recomputed backward fields, residuals, and vector field.

Compare actual finite dense gradient flow to this proxy interpolant, stopping it only if it leaves a slightly larger physical tube. The identical reference inequality and integral comparison give

$$\sup_{t \leq T} d_n(\theta_{n,D}(t), \theta_{n,\text{proxy}}^h(t)) \leq C_T e^{C(1+R)T} \{(1+R)h + e^{-cR^2} + o_{\mathbb{P}}(1)\}.$$

The last term is at fixed h, R . The proof uses empirical tails only of fixed-program reference nodes. Taking width to infinity at fixed h, R , then h to zero, then R to infinity makes the error smaller than the exit margin and excludes the stop. Combining this with (44) proves the compact-time observable convergence. For the original $C^{1,1}$ activations insert a fixed smooth reference program and its $C(1+R)\epsilon$ comparison error, taking its smoothing scale to zero before the final R limit. Passive test inputs require only finitely many extra forward queries and first-layer root coordinates; they change no training update. They are therefore covered by exactly the same fixed-program and comparison arguments.

We spell out how this yields actual dense carrier tails. For fixed M , use the continuous 1-Lipschitz function $g_M(v) = (|v| - M)_+$, with $|v| \mathbf{1}_{\{|v| > 2M\}} \leq 2g_M(v)$. Its normalized Euclidean norm changes by at most the RMS change in the carrier. Fixed-program convergence transfers its squared norm at every reference node. A finite time grid, the uniform parameter-speed bound, and (42) against

these reference nodes control the interpolation error. First choose the grid and a comparison cutoff, let width tend to infinity at that fixed program, then refine the grid and remove the comparison cutoff. The population Gaussian bound implies a limiting carrier cutoff norm at most Ce^{-cM^2} , after halving M and changing c . Prediction convergence also transfers the residual factor. We conclude that for every fixed $M \geq 1$ and $\epsilon > 0$,

$$\Pr \left\{ \int_0^T \rho_D(t) H_n(M, t) dt > Ce^{-cM^2} + \epsilon \right\} \longrightarrow 0.$$

The constant is independent of T : population carrier tails are uniform in time and total residual activity is bounded. On $\mathcal{G}_n$, finite dense RMS bounds imply $H_n(M, t) \leq C$, and (32) gives $\int_T^\infty \rho_D H_n dt \leq Ce^{-\kappa T}$ uniformly in n, M . Choosing T after the error tolerance but before the width limit proves

$$\Pr\{Z_n(M) > Ce^{-cM^2} + \epsilon\} \longrightarrow 0 \quad \text{for every fixed } M \geq 1, \epsilon > 0.$$

Set

$$a_n = \sup_{M \in \mathbb{N}, M \geq 1} (Z_n(M) - Ce^{-cM^2})_+.$$

These random variables are measurable, bounded, and depend only on the dense flow and its initialization. To prove $a_n \rightarrow 0$, fix a tolerance and choose one integer J with Ce^{-cJ^2} smaller than half that tolerance. Monotonicity of Z_n bounds every excess at $M \geq J$ by $Z_n(J)$, which is small in probability. The finitely many smaller integers are handled by a union bound and fixed-cutoff convergence. This proves (45) simultaneously in M , with no width-dependent cutoff theorem and no numerical width rate. $\square$

A.4 Damping and near-quadratic tracking

We now compare the actual autonomous closure to the dense flow. All constants in this subsection are independent of width, order and physical time. Fix an initialization in (20). Denote the full dense backward carriers by $k_{D,a}^{(L)} = w_D$ and $k_{D,a}^{(\ell)} = (W_D^{(\ell+1)})^\top \delta_{D,a}^{(\ell+1)}$ for $\ell < L$. The preceding Gaussian argument gives, simultaneously at integer cutoffs $M \geq 1$,

$$\begin{aligned} H_n(M, t) &= \sum_{\ell=1}^L \max_a \frac{\|k_{D,a}^{(\ell)}(t) \mathbf{1}_{|k_{D,a}^{(\ell)}(t)| > M}\|_2}{\sqrt{n}}, \\ Z_n(M) &= \int_0^\infty \rho_D(t) H_n(M, t) dt \leq Ce^{-cM^2} + a_n, \quad a_n \xrightarrow{\Pr} 0. \end{aligned} \quad (46)$$

Here the indicator acts coordinatewise, and a_n depends only on the dense reference. The physical bounds imply $H_n \leq C$. Put

$$D = \sup_{t \geq 0} d_n(\hat{\theta}(t), \theta_D(t)), \quad \epsilon = \int_0^\infty e_E(t) dt, \quad Q_r = \int_0^\infty \|\hat{r}(t) - r_D(t)\|_m dt.$$

All three quantities are finite before making any comparison, by the all-order fitting and speed bounds. They depend on n, q ; this dependence is suppressed only in this proof.

One-reference damping. Forward subtraction through the bounded operators and slopes gives

$$\max_{\ell, a} \frac{\|\hat{z}_a^{(\ell)} - z_{D,a}^{(\ell)}\|_2 + \|\hat{h}_a^{(\ell)} - h_{D,a}^{(\ell)}\|_2}{\sqrt{n}} \leq C d_n(t).$$

Indeed, at each hidden link subtract as $(\widehat{W} - W_D)h_D + \widehat{W}(\widehat{h} - h_D)$ and use the feature RMS and operator bounds. In backward subtraction the only problematic product is a changed gate multiplied by a dense carrier. Splitting at M gives

$$\frac{\| [g_\ell(\widehat{z}) - g_\ell(z_D)] \odot k_D \|_2}{\sqrt{n}} \leq jM \frac{\|\widehat{z} - z_D\|_2}{\sqrt{n}} + 2s \frac{\|k_D \mathbf{1}_{|k_D| > M}\|_2}{\sqrt{n}}. \quad (47)$$

The other terms are a bounded gate times $\widehat{W}^\top(\widehat{\delta} - \delta_D)$ and a bounded gate times $(\widehat{W} - W_D)^\top \delta_D$. At the top the first term is instead the readout difference. Descending through the fixed depth therefore gives

$$\max_{\ell, a} \frac{\|\widehat{\delta}_a^{(\ell)} - \delta_{D,a}^{(\ell)}\|_2}{\sqrt{n}} \leq C[(1+M)d_n(t) + H_n(M, t)]. \quad (48)$$

The factor M is added at each layer; it is not multiplied by another cutoff when propagating the next backward difference.

Expanding the products in the tangent Gram (31), Cauchy–Schwarz yields the same bound for $\|\widehat{\Gamma} - \Gamma_D\|_{\text{op}}$. The sample factor $1/m$ makes a matrix whose entries are bounded by B/m have operator norm at most B . For $u = \widehat{r} - r_D$ the exact residual difference equation is

$$\dot{u} = -2\widehat{\Gamma}u - 2(\widehat{\Gamma} - \Gamma_D)r_D + \widehat{J}E.$$

The preserved Gram gap and $\|\widehat{J}E\|_m \leq CYe_E$ imply

$$D^+ \|u\|_m \leq -\lambda \|u\|_m + C\rho_D[(1+M)d_n + H_n] + Ce_E.$$

At zeros of u this inequality follows by differentiating $\sqrt{\|u\|_m^2 + \zeta^2}$ and then letting $\zeta \downarrow 0$. Integrating and dropping the nonnegative terminal norm gives

$$\int_0^t \|u\|_m ds \leq C \left[(1+M) \int_0^t \rho_D d_n ds + Z_n(M; t) + \epsilon(t) \right], \quad (49)$$

where a final argument t means that an integral stops at t .

Subtract the readout updates as $u\widehat{h} + r_D(\widehat{h} - h_D)$. In a hidden update use the three differences of the residual, backward response and forward response. The rank-one Frobenius identity and (48) give

$$d_n(t) \leq C\epsilon(t) + C \int_0^t \|u\|_m ds + C \int_0^t \rho_D[(1+M)d_n + H_n] ds.$$

Insert (49). On each terminal interval, replace its nondecreasing source by its terminal value and apply the integrating factor for $C(1+M)\rho_D$. Since $\int_0^\infty \rho_D \leq CY$, then taking the supremum gives

$$D \leq A_M(\epsilon + Z_n(M)), \quad A_M = Ce^{KM} \geq 1, \quad Q_r \leq C[(1+M)D + Z_n(M) + \epsilon]. \quad (50)$$

Only the dense reference supplied the carrier tail in this argument.

A smoother history used only in the proof. Let clip_M act coordinatewise by truncation to $[-M, M]$. At the actual closure state define auxiliary backward fields

$$\begin{aligned} \delta_a^{(L),M} &= g_L(\widehat{z}_a^{(L)}) \odot \text{clip}_M(\widehat{w}), \\ \delta_a^{(\ell),M} &= g_\ell(\widehat{z}_a^{(\ell)}) \odot \text{clip}_M((\widehat{W}^{(\ell+1)})^\top \delta_a^{(\ell+1),M}). \end{aligned} \quad (51)$$

These fields are not evolved or used by the closure. Clipping contracts Euclidean distance and decreases magnitudes, so their RMS norms are at most CY . At the top, the almost-everywhere chain rule gives

$$\frac{\|\dot{\delta}_a^{(L),M}\|_2}{\sqrt{n}} \leq jM \frac{\|\dot{\hat{z}}_a^{(L)}\|_2}{\sqrt{n}} + s \frac{\|\dot{w}\|_2}{\sqrt{n}}.$$

At a lower layer the corresponding upper bound is

$$jM \frac{\|\dot{\hat{z}}_a^{(\ell)}\|_2}{\sqrt{n}} + s \|\dot{\hat{W}}^{(\ell+1)}\|_{\text{op}} \frac{\|\delta_a^{(\ell+1),M}\|_2}{\sqrt{n}} + s \|\widehat{W}^{(\ell+1)}\|_{\text{op}} \frac{\|\dot{\delta}_a^{(\ell+1),M}\|_2}{\sqrt{n}}.$$

Lipschitz functions composed with absolutely continuous coordinates are absolutely continuous, so this calculation also applies when g_ℓ or clipping is not classically differentiable at every point. Downward induction and (33) give $\|\delta_a^{(\ell),M}\|_2/\sqrt{n} \leq C(1+M)\hat{\rho}$. Clipping the readout is essential here: unbounded activations give an RMS bound on w , not a width-independent coordinate bound.

Set $c = \hat{r}/\hat{\rho}$ and $b_a^{(\ell),M} = c_a \delta_a^{(\ell),M}$, extended by zero over the prefix. The zero readout makes the join continuous. Since $\|\dot{c}\|_m \leq C$ and m is fixed,

$$\frac{\|\dot{b}_a^{(\ell),M}(t)\|_2}{\sqrt{n}} \leq C[1 + (1+M)\hat{\rho}(t)]. \quad (52)$$

Freeze this proof history after physical time T ; denote the frozen history by b_T^M . Its difference from b^M has normalized clock L^2 norm at most $Ce^{-\kappa T/2}$, because its amplitude is bounded and its support has length at most $\int_T^\infty \hat{\rho}$. For any current endpoint $A = \tau(t)$, the weighted derivative energy obeys

$$\begin{aligned} \frac{1}{n} \int_0^A \xi(A-\xi) \| (b_T^M)'(\xi) \|_2^2 d\xi &\leq \frac{\sup_t \tau(t)}{\kappa n} \int_0^{\min(t,T)} \|\dot{b}^M(s)\|_2^2 ds \\ &\leq C[T + (1+M)^2]. \end{aligned} \quad (53)$$

Indeed $A - \tau(s) \leq \int_s^\infty \hat{\rho} \leq \hat{\rho}(s)/\kappa$ cancels the inverse clock speed. The remaining integral is bounded using (52) and $\int_0^\infty \hat{\rho}^2 < \infty$. On each finite interval before freezing the residual is positive, so the frozen history is in H^1 . The weighted Legendre estimate (21) and projection contraction now give

$$\left[\frac{1}{n} \int_0^A \|(I - \Pi_q^A) b_a^{(\ell),M}\|_2^2 d\xi \right]^{1/2} \leq C \left[\frac{1+M+\sqrt{T}}{q} + e^{-\kappa T/2} \right]. \quad (54)$$

Returning to the actual backward history. Compare (51) with the dense backward recursion. At the top, insert the dense clipped readout: clipping contraction controls the readout difference, the changed gate costs CMd_n , and the omitted dense tail is a term of H_n . At every lower layer the clipped-input difference is bounded by the next auxiliary/dense backward difference plus $CY\|\widehat{W} - W_D\|_F$. The changed gate and clipping tail have the same bounds as at the top. Induction, and then (48), yield

$$\max_{\ell,a} \frac{\|\widehat{\delta}_a^{(\ell)} - \delta_a^{(\ell),M}\|_2}{\sqrt{n}} \leq C[(1+M)d_n + H_n].$$

Multiplying by the same c_a and integrating in the closure's own clock gives

$$\left[\frac{1}{n} \int_0^A \|b_a^{(\ell)} - b_a^{(\ell),M}\|_2^2 d\xi \right]^{1/2} \leq C(1+M)D + C\sqrt{Z_n(M) + Q_r}. \quad (55)$$

For the tail term, use $H_n \leq C$ and $|\hat{\rho} - \rho_D| \leq \|\hat{r} - r_D\|_m$ to obtain $\int \hat{\rho} H_n^2 \leq C Z_n(M) + C Q_r$. There is no assumed bound on the ratio of the two residuals.

The forward history tail is at most C/q by (29) and (21). Use this factor, (54), (55), and the exact absolute-defect inequality (28). Increasing its terminal time to infinity gives

$$\epsilon \leq C \left[\frac{1 + M + \sqrt{T}}{q^2} + \frac{e^{-\kappa T/2}}{q} + \frac{(1 + M)D}{q} + \frac{\sqrt{Z_n(M) + Q_r}}{q} \right]. \quad (56)$$

The bounds are uniform in the terminal time and the accumulated absolute defect increases to ϵ , which justifies this limit.

Absorbing the remaining feedback. Put $U = \epsilon + Z_n(M)$. By (50), $D \leq A_M U$ and $Z_n(M) + Q_r \leq C(1 + M)A_M U$. Consequently

$$U \leq Z_n(M) + C \left[\frac{1 + M + \sqrt{T}}{q^2} + \frac{e^{-\kappa T/2}}{q} \right] + \frac{C(1 + M)A_M U}{q} + \frac{C\sqrt{(1 + M)A_M}}{q} \sqrt{U}.$$

If $C(1 + M)A_M/q \leq 1/4$, absorb the linear term and apply $h\sqrt{U} \leq U/4 + h^2$ to the last term. Multiplication by A_M gives

$$D \leq C A_M Z_n(M) + C A_M \left[\frac{1 + M + \sqrt{T}}{q^2} + \frac{e^{-\kappa T/2}}{q} \right] + \frac{C(1 + M)A_M^2}{q^2}. \quad (57)$$

Set $u = q^{-2} + a_n$. For small u , choose an integer $M \leq C + C\sqrt{\log(e + 1/u)}$ making $Ce^{-cM^2} \leq u$, and $T \leq C\log(e + 1/u)$ making $e^{-\kappa T/2} \leq \sqrt{u}$. Then $Z_n(M) \leq Cu$. Since $u \geq q^{-2}$, the cutoff is at most $C + C\sqrt{\log(e + q)}$, so the absorption condition holds above one order threshold independent of width and of a_n . Each term of (57) is bounded by

$$C\Phi(q^{-2} + a_n), \quad \Phi(u) = u \exp\{K\sqrt{\log(e + 1/u)}\}, \quad \Phi(0) = 0. \quad (58)$$

Enlarging K absorbs the logarithmic factors. The common physical bound on D handles the finitely many smaller orders and nonsmall u . Finally $\Phi(u)/u$ is decreasing, hence $\Phi(v + w) \leq \Phi(v) + \Phi(w)$: multiply its value at $v + w$ separately by v and w . Set $b_n = C\Phi(a_n)$ and enlarge K once more to bound $\Phi(q^{-2})$ by $\omega(q)$. This proves (14). Continuity of Φ at zero gives $b_n \rightarrow 0$ in probability, uniformly in the choice of order.

A.5 Whole-input prediction and population convergence

The full first-layer Frobenius bound in (20), together with (33), controls $\|W^{(1)}\|_F/\sqrt{n}$ for both paths at every time. Linear activation growth and forward induction give, for any $x \in \mathbb{R}^d$,

$$\begin{aligned} \frac{\|h^{(\ell)}(t, x)\|_2}{\sqrt{n}} &\leq C(1 + \|x\|_2/\sqrt{d}), \\ |\hat{f}(t, x) - f_D(t, x)| &\leq C(1 + \|x\|_2/\sqrt{d})d_n(t). \end{aligned} \quad (59)$$

For the second inequality, subtract the forward recursions as before and expand the readout pairing into readout difference and feature difference. The constant term is needed for activations with nonzero offsets. Taking the time supremum before integrating proves the closure-to-dense version of (15); taking a supremum on a bounded input set proves its absolute-error counterpart.

To identify the dense target on all inputs, add the first-layer Gaussian row coordinates and any finite collection of input probes to the dense reference programs. These are passive observations and never enter the training residuals. The finite-program limit and finite-time reference comparison already proved give convergence in probability of their predictions on every compact time interval. The global slope and weight bounds imply, for both dense and closure predictors,

$$\begin{aligned} |f(t, x) - f(t, x')| &\leq C\|x - x'\|_2/\sqrt{d}, \\ |\partial_t f(t, x)| &\leq Ce^{-\kappa t}(1 + \|x\|_2/\sqrt{d}). \end{aligned} \quad (60)$$

For the second inequality, differentiate the forward recursion using (33), then differentiate the final readout pairing. The bounds hold uniformly in n, q on the common events.

A finite net of any bounded input set and the first inequality extend dense probe convergence to the entire set. The integrated second inequality bounds the remaining variation after time T by $Ce^{-\kappa T}(1 + \|x\|_2/\sqrt{d})$, uniformly in width. Choose T large, then take the width limit on $[0, T]$. This proves all-time dense prediction convergence uniformly on bounded input sets to the uniquely constructed dense population predictor f_∞ . For a general fixed test law with finite second moment, truncate to a bounded input set. The complement contributes at most a constant times $\int_{\|x\|>R}(1 + \|x\|_2/\sqrt{d})^2 d\mu$, which tends to zero. Thus $d_{n,\mu} := \mathcal{E}_\mu(f_{n,D}, f_\infty) \rightarrow 0$ in probability. The triangle inequality and (59) give (15) with $b_{n,\mu} = C_\mu b_n + d_{n,\mu}$, independent of q . This completes the proof of Theorem 3.1.

For precision about fixed-order limits, compactify time by $u = e^{-\kappa t} \in [0, 1]$. Equations (59)–(60) give uniform boundedness and equicontinuity on every compact input set, including the endpoint $u = 0$. Finite nets and a diagonal selection over integer input radii prove compactness in local uniform convergence. The same second-moment majorant upgrades this to the test norm (11). On events of probability tending to one this gives tightness of each fixed-order predictor family. Every subsequential limit law inherits $\mathcal{E}_\mu(F_q, f_\infty) \leq C_\mu \omega(q)$ almost surely: for any positive margin the finite-width violation probability tends to zero. This identifies accuracy of predictor limit points, without asserting uniqueness of their law or of a population moment-state ODE.

B Finite-horizon tracking without small labels

The learning-speed closure also tracks arbitrary bounded labels on a prescribed finite horizon. The constant in the resulting q^{-1} bound is independent of width, but the sufficient order depends on width. The theorem below specifies that dependence and proves continuation of the closure; it assumes neither a Gram gap nor successful fitting.

For parameter increments use the mobility norm

$$|v|_n^2 = \frac{\|v_1\|_F^2}{n} + \sum_{\ell=2}^L \|v_\ell\|_F^2 + \frac{\|v_w\|_2^2}{n}. \quad (61)$$

The paper's distance d_n , the sum of the normalized block norms, satisfies

$$|\theta - \vartheta|_n \leq d_n(\theta, \vartheta) \leq \sqrt{L+1} |\theta - \vartheta|_n.$$

Write $\|u\|_m^2 = m^{-1} \sum_a u_a^2$ for sample RMS and $X = \max_a \|x_a\|_2/\sqrt{d}$.

Proposition B.1 (Finite horizons and a joint width–order bound). *Fix finite data inputs, $L \geq 2$, and constants $K, A_0 < \infty$. The layer activations satisfy*

$$\phi_\ell \in C_{\text{loc}}^{1,1}(\mathbb{R}), \quad a_\ell = |\phi_\ell(0)| < \infty, \quad v_\ell = \|\phi'_\ell\|_\infty < \infty.$$

At every width n , allow any finite initialization and labels satisfying

$$\max_{\ell \geq 2} \|W_0^{(\ell)}\|_{\text{op}} \leq K, \quad \max_a \frac{\|W_0^{(1)} x_a / \sqrt{d}\|_2}{\sqrt{n}} \leq A_0, \quad B_0 := \frac{\|w_0\|_2}{\sqrt{n}} \leq 1, \quad Y := \|y\|_m \leq 1. \quad (62)$$

Let θ_D be the canonical dense flow and $\hat{\theta}_{n,q}$ the original order- q learning-speed closure from the same initialization. For each $T < \infty$ there are $R_T, C_T, \Lambda_T \geq 1$, depending only on T , the fixed data inputs, L, K, A_0 , and $(a_\ell, v_\ell)_{\ell=1}^L$, with the following property. Set

$$\ell_{n,T} = \max_\ell \text{Lip}(\phi'_\ell; [-R_T\sqrt{n}, R_T\sqrt{n}]), \quad s_{n,T} = 1 + \sqrt{n} \ell_{n,T}, \quad H_{n,T} = \exp(\Lambda_T s_{n,T}). \quad (63)$$

For every width $n \geq 1$ and every integer

$$q \geq \max\{B_0^2 H_{n,T}^2, s_{n,T} H_{n,T}\}, \quad (64)$$

the closure exists uniquely on $[0, T]$ and

$$\sup_{t \leq T} d_n(\hat{\theta}_{n,q}(t), \theta_D(t)) \leq C_T H_{n,T} \left(\frac{B_0}{q^{3/2}} + \frac{s_{n,T}}{q^2} \right) \leq \frac{2C_T}{q}. \quad (65)$$

Thus $q \geq \lceil e^{2\Lambda_T s_{n,T}} \rceil$ is sufficient in general; if $w_0 = 0$, it suffices that $q \geq \lceil s_{n,T} e^{\Lambda_T s_{n,T}} \rceil$. The bounds $Y, B_0 \leq 1$ can be replaced by any fixed finite bounds, with corresponding changes in the constants.

For globally Lipschitz gates, $\ell_{n,T} \leq J$ for a fixed J , so a sufficient order has $\log q \geq C_T(1 + \sqrt{n})$. This is a bound on a joint region of (n, q) , rather than a C_T/q estimate for unrestricted widths and orders. It also imposes no smallness requirement on labels relative to a Gram margin. The large sufficient orders may exceed n/m , so this conclusion alone does not guarantee a reduction in matrix rank or moving state.

The unchanged autonomous system. For completeness, let $p_k(u) = P_k(2u - 1)$, where P_k is the Legendre polynomial normalized by $P_k(1) = 1$. For each hidden link and sample the closure stores q moments $\bar{h}_{a,k}^{(\ell-1)}$ and $\bar{\delta}_{a,k}^{(\ell)}$. Its clock satisfies $\dot{\tau} = \rho$, $\tau(0) = 1$, and either moment M_k obeys

$$\dot{M}_k = S - \frac{\rho}{\tau} \left[k M_k + \sum_{j < k} (2j + 1) M_j \right], \quad 0 \leq k < q, \quad (66)$$

with source $S = \rho h_a^{(\ell-1)}$ or $S = r_a \delta_a^{(\ell)}$. Only the degree-zero forward moments are initially nonzero, equal to the initial features. The hidden matrices are

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a,k < q} (2k + 1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top. \quad (67)$$

All fields in these equations come from the reconstructed network; the first layer and readout follow the canonical dense update formulas. The raw system is locally Lipschitz at fixed n, q , including at $\rho = 0$, because it uses ρ and $r_a \delta_a$, and divides only by $\tau > 0$. If $\rho(0) = 0$, every raw velocity vanishes and both paths are stationary. Otherwise backward uniqueness prevents a regular solution from reaching such an equilibrium in finite time. Thus the clock change used below is legitimate on compact regular intervals.

The proof first obtains a uniform region from dense loss dissipation and stops the closure at distance one from the dense path. Forward-history energy then controls the defect relative to the residual, before any backward derivative estimate is used. The resulting backward regularity supplies a second projection factor. An explicit stability estimate and a strict distance bound finally remove the stop.

Polynomial estimates and the exact defect

Let Π_q be the $L^2(0, \tau)$ orthogonal projection onto polynomials of degree below q , acting on any finite-dimensional Hilbert space $\mathcal{H}$. The following estimates have dimension-independent constants.

Lemma B.2 (Legendre tails and endpoint bounds).

$$\|(I - \Pi_q)g\|_{L^2}^2 \leq \frac{1}{q(q+1)} \int_0^\tau \xi(\tau - \xi) \|g'(\xi)\|_{\mathcal{H}}^2 d\xi \leq \frac{\tau^2 \|g'\|_{L^2}^2}{4q(q+1)}, \quad (68)$$

$$\|g(\tau) - (\Pi_q g)(\tau)\|_{\mathcal{H}} \leq C\sqrt{\tau/q} \|g'\|_{L^2}, \quad \|(\Pi_q b)(\tau)\|_{\mathcal{H}} \leq C\sqrt{q} \|b\|_{L^\infty(\mathcal{H})}, \quad (69)$$

$$\|(I - \Pi_q)\mathbf{1}_{[1, \tau]}\|_{L^2(0, \tau)} \leq C\sqrt{\tau/q} \quad (\tau \geq 1). \quad (70)$$

The first two estimates concerning g require $g \in H^1$; the estimate for b requires only boundedness.

Proof. The weighted tail and endpoint estimates are those of Lemma A.1, with $A = \tau$; its proof applies to the same Hilbert-valued histories. To prove the step estimate, for $q \geq 2$ replace the step in (70) by a linear ramp of width τ/q on a side of its jump with that much room. One side always has length at least $\tau/2$. The replacement error is at most $\sqrt{\tau/q}$ and the derivative norm is $\sqrt{q/\tau}$. Projection contraction and (68) prove the claim. For $q = 1$ contraction suffices, and at $\tau = 1$ the step vanishes almost everywhere. $\square$

On the closure's clock use the constant forward prefix on $[0, 1]$ and the zero backward prefix, followed by the actual histories $h_a^{(\ell)}$ and $b_a^{(\ell)} = (r_a/\rho)\delta_a^{(\ell)}$. The moment integrals of these histories obey (66), by the polynomial identity $up'_k = kp_k + \sum_{j < k} (2j+1)p_j$; uniqueness therefore identifies them with the raw moments. A star denotes projection at the current endpoint. For $D_g = \int_0^\tau \|g - \Pi_q g\|_2^2 d\xi$,

$$\dot{D}_g = \rho \|g - g^*\|_2^2, \quad D_g(0) = 0. \quad (71)$$

Indeed, the derivative of the fitted polynomial lies in the polynomial space, so its pairing with the error is zero. Alternatively, in a fixed monomial basis with Gram matrix $G(\tau)$, the projected pairing of b, h is $M_b G^{-1} M_h^\top$. Differentiation using $M'_b = b(\tau)v(\tau)^\top$ and $G' = v(\tau)v(\tau)^\top$ gives $b(h^*)^\top + b^*h^\top - b^*(h^*)^\top$. Applying this identity to (67) yields exactly

$$\hat{\theta} = F(\hat{\theta}) + E, \quad E_1 = E_w = 0, \quad E_\ell = \frac{2\rho}{mn} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top. \quad (72)$$

These differentiation formulas require only locally integrable histories; the backward prefix jump causes no extra term. The rank-one identity $\|uv^\top/n\|_F = (\|u\|_2/\sqrt{n})(\|v\|_2/\sqrt{n})$ and Cauchy-Schwarz in (71) give

$$\int_0^t \|E_\ell\|_F ds \leq \frac{2}{mn} \sum_a \sqrt{D_{b, \ell, a}(t) D_{h, \ell-1, a}(t)}. \quad (73)$$

A uniform comparison region and forward regularity

Write $\eta = \rho(0)$. Since $|\phi_\ell(u)| \leq a_\ell + v_\ell|u|$, the recursion $M_1^0 = a_1 + v_1 A_0$, $M_\ell^0 = a_\ell + v_\ell K M_{\ell-1}^0$ gives $\eta \leq Q_0 := 1 + M_L^0$. Dense loss dissipation in the mobility norm is

$$\frac{d}{dt} \rho_D^2 = -|F(\theta_D)|_n^2, \quad |\theta_D(t) - \theta_0|_n \leq \eta\sqrt{t}.$$

The second inequality is Cauchy–Schwarz applied to the integrated first one. It bounds every dense coordinate on finite intervals at fixed width, and the locally Lipschitz dense vector field consequently extends past any proposed finite maximal time.

Stop the closure at its first mobility distance $|\hat{\theta} - \theta_D|_n = 1$, or at its maximal existence time, within $[0, T]$. Put $D_T = Q_0\sqrt{T}$. Up to this stop both paths, and every parameter segment joining them, belong to the convex region

$$\begin{aligned} \|W^{(\ell)}\|_{\text{op}} &\leq D := K + D_T + 1 \quad (\ell \geq 2), & \frac{\|w\|_2}{\sqrt{n}} &\leq B := 2 + D_T, \\ \max_a \frac{\|W^{(1)}x_a/\sqrt{d}\|_2}{\sqrt{n}} &\leq A := A_0 + X(D_T + 1). \end{aligned} \tag{74}$$

Define

$$M_1 = a_1 + v_1 A, \quad M_\ell = a_\ell + v_\ell D M_{\ell-1}, \quad \beta_L = v_L B, \quad \beta_\ell = v_\ell D \beta_{\ell+1}.$$

Throughout this region, the feature and backward RMS norms at each sample are at most M_ℓ and β_ℓ , respectively. Choose

$$R_T = 1 + \max\{A, D M_1, \dots, D M_{L-1}\}.$$

Every preactivation coordinate in the entire comparison region belongs to $[-R_T\sqrt{n}, R_T\sqrt{n}]$, justifying (63) without using a local modulus to construct its own domain. Also $\rho \leq \rho_* := 1 + B M_L$, and $1 \leq \tau \leq A_c := 1 + T \rho_*$. All these constants are independent of n, q and of gate moduli.

In estimates for the closure we suppress hats and use the sample/neuron Hilbert norm $\|g\|_{\mathcal{H}}^2 = (mn)^{-1} \sum_a \|g_a\|_2^2$. Let $Z_\ell = \int_0^\tau \|\partial_\xi h^{(\ell)}\|_{\mathcal{H}}^2 d\xi$ and $e_E = \sum_{\ell=2}^L \|E_\ell\|_F$. All forward histories belong to H^1 on compact pre-stop intervals: their physical curves are absolutely continuous, ρ is positive there, and their prefixes join continuously. The backward array satisfies $\|b^{(\ell)}\|_{\mathcal{H}} \leq \beta_\ell$. Endpoint evaluation on degree-below- q polynomials has norm $q/\sqrt{\tau}$, since $\sum_{k < q} (2k+1) = q^2$. Projection contraction therefore bounds its endpoint error by $(q+1)\beta_\ell$. Inserting this into (72), and using (71) and (68), gives

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 ds \leq 4(q+1)^2 \beta_\ell^2 \left\| (I - \Pi_q) h^{(\ell-1)} \right\|_{L^2(\mathcal{H})}^2 \leq 2A_c^2 \beta_\ell^2 Z_{\ell-1}. \tag{75}$$

The first-layer clock derivative has RMS at most $2v_1 X^2 \beta_1$. For later layers differentiate the forward recurrence:

$$\partial_\xi h_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + W^{(\ell)} \partial_\xi h_a^{(\ell-1)} \right].$$

Here $\|F_\ell/\rho\|_F \leq 2\beta_\ell M_{\ell-1}$. The three-term squared triangle inequality and (75) prove inductively $Z_\ell \leq Z_\ell^*$, where, with $S = T \rho_*$,

$$\begin{aligned} Z_1^* &= 4S v_1^2 X^4 \beta_1^2, \\ Z_\ell^* &= 3v_\ell^2 \left[4S \beta_\ell^2 M_{\ell-1}^4 + (D^2 + 2A_c^2 \beta_\ell^2 M_{\ell-1}^2) Z_{\ell-1}^* \right]. \end{aligned}$$

Thus forward derivative energy is bounded independently of width and order. By (69), the forward endpoint error is at most $C_T q^{-1/2}$ and the backward endpoint error at most $C_T q^{1/2}$, in $\mathcal{H}$. Their product in (72) proves

$$e_E \leq C_T \rho, \quad |\dot{\hat{\theta}}|_n \leq C_T \rho, \quad \max_{\ell, a} \frac{\left\| \dot{z}_a^{(\ell)} \right\|_2 + \left\| \dot{h}_a^{(\ell)} \right\|_2}{\sqrt{n}} \leq C_T \rho. \tag{76}$$

The last estimate follows by forward differentiation through bounded operators and slopes. The same calculation for an arbitrary parameter increment v , including $Df_a[v] = v_w^\top h_a^{(L)}/n + w^\top Dh_a^{(L)}[v]/n$, gives $\|Df[v]\|_m \leq C_T|v|_n$ throughout (74). Consequently $\|\dot{r}\|_m \leq C_T\rho$ and $|\dot{\rho}| \leq C_T\rho$. For $\eta > 0$ this proves

$$c_T\eta \leq \rho(t) \leq C_T\eta, \quad \int_0^t \rho \, ds \leq C_T\eta, \quad \|\dot{c}\|_m \leq C_T \quad (c = r/\rho), \quad Z_\ell(t) \leq C_T\eta. \quad (77)$$

The third bound uses $\dot{c} = (\dot{r} - c\dot{\rho})/\rho$ and $\|c\|_m = 1$; the last uses (76) and $d\xi = \rho \, dt$. The positive lower bound is derived relative to η and is not an additional hypothesis.

The second projection factor and stability

Integrating the readout equation and using (77) refines the backward bounds to

$$\frac{\|w(t)\|_2}{\sqrt{n}} \leq B_0 + C_T\eta, \quad \max_{\ell,a} \frac{\|\delta_a^{(\ell)}(t)\|_2}{\sqrt{n}} \leq C_T(B_0 + \eta). \quad (78)$$

Use the full carriers $k_a^{(L)} = w$ and $k_a^{(\ell)} = (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}$. Their RMS norms are bounded on (74), so $\|k_a^{(\ell)}\|_\infty \leq \|k_a^{(\ell)}\|_2 \leq C_T\sqrt{n}$. A Lipschitz scalar function composed with an absolutely continuous curve is absolutely continuous, with derivative bounded almost everywhere by its Lipschitz constant times the curve speed. This follows directly from the disjoint-interval criterion for absolute continuity and then from difference quotients. Applying it to ϕ'_ℓ gives

$$\left| \frac{d}{dt} \phi'_\ell(z_{a,i}^{(\ell)}) \right| \leq \ell_{n,T} |\dot{z}_{a,i}^{(\ell)}| \quad \text{a.e.}$$

Differentiate $\delta_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot k_a^{(\ell)}$. Its gate-variation term has RMS at most $C_T\sqrt{n}\ell_{n,T}\rho$ by (76). For $\ell < L$,

$$\dot{k}_a^{(\ell)} = (\dot{W}^{(\ell+1)})^\top \delta_a^{(\ell+1)} + (W^{(\ell+1)})^\top \dot{\delta}_a^{(\ell+1)}.$$

The first term has RMS at most $C_T\rho$ and the second propagates the next derivative with coefficient D . Starting with the readout derivative proves, through fixed depth,

$$\max_{\ell,a} \frac{\|\dot{\delta}_a^{(\ell)}\|_2}{\sqrt{n}} \leq C_T s_{n,T}\rho, \quad \|\dot{b}^{(\ell)}\|_{\mathcal{H}} \leq C_T(B_0 + s_{n,T}\rho).$$

For the last bound use $b = c\delta$, (78), $\|\dot{c}\|_m \leq C_T$, and $\eta \leq C_T\rho$. Only the full-carrier RMS-to-maximum conversion introduces width here; no tail assumption on a dense carrier is used.

The initial backward jump $b_0^{(\ell)} = b^{(\ell)}(1+)$ has $\mathcal{H}$ norm at most C_TB_0 . Subtract $b_0^{(\ell)}\mathbf{1}_{[1,\tau]}$ from its history to obtain a continuous remainder $\tilde{b}^{(\ell)}$. Changing variables and using (77) gives

$$\int_0^\tau \|\partial_\xi \tilde{b}^{(\ell)}\|_{\mathcal{H}}^2 \, d\xi = \int_0^t \frac{\|\dot{b}^{(\ell)}\|_{\mathcal{H}}^2}{\rho} \, ds \leq C_T \left(\frac{B_0^2}{\eta} + s_{n,T}^2 \right).$$

If Q_h, Q_b denote the corresponding $L^2(\mathcal{H})$ projection tails, (68), (70), and (77) yield

$$Q_{h,\ell} \leq \frac{C_T\sqrt{\eta}}{q}, \quad Q_{b,\ell} \leq C_T \left[\frac{B_0}{\sqrt{q}} + \frac{B_0/\sqrt{\eta} + s_{n,T}\sqrt{\eta}}{q} \right].$$

Sample Cauchy–Schwarz in (73) gives $\int_0^t \|E_\ell\|_F ds \leq 2Q_{b,\ell}Q_{h,\ell-1}$. Multiplying the tails before discarding their residual factors proves

$$\int_0^t e_E ds \leq C_T \left[\frac{B_0\sqrt{\eta}}{q^{3/2}} + \frac{B_0 + s_{n,T}\eta}{q^2} \right] \leq C_T \left[\frac{B_0}{q^{3/2}} + \frac{s_{n,T}}{q^2} \right]. \quad (79)$$

The inverse residual has canceled. Thus the constants stay uniform as $\eta \downarrow 0$; the exactly zero case was separated before division. When $w_0 = 0$, there is no prefix jump and no $q^{-3/2}$ term.

To compare two parameters θ, ϑ in (74), forward subtraction gives

$$\max_{\ell,a} \frac{\left\| \Delta z_a^{(\ell)} \right\|_2 + \left\| \Delta h_a^{(\ell)} \right\|_2}{\sqrt{n}} \leq C_T |\theta - \vartheta|_n.$$

For example $\Delta z^{(\ell)} = \Delta W^{(\ell)} h^{(\ell-1)}(\vartheta) + W^{(\ell)}(\theta) \Delta h^{(\ell-1)}$; the two terms are controlled by the feature RMS and the operator norm, respectively. In backward subtraction put the full carrier of ϑ in the gate-difference term. Its maximum norm is at most $C_T \sqrt{n}$, so that term has RMS at most $C_T \sqrt{n} \ell_{n,T} |\theta - \vartheta|_n$. The matrix-difference term is bounded by its Frobenius norm times a bounded backward RMS, and the next backward difference propagates through bounded operators and slopes. Descending from the readout gives

$$\max_{\ell,a} \frac{\left\| \Delta \delta_a^{(\ell)} \right\|_2}{\sqrt{n}} \leq C_T s_{n,T} |\theta - \vartheta|_n.$$

Subtracting the canonical gradient formulas one factor at a time now proves

$$|F(\theta) - F(\vartheta)|_n \leq C_T s_{n,T} |\theta - \vartheta|_n. \quad (80)$$

This finite-difference argument uses locally Lipschitz gates and does not assume an everywhere-defined second activation derivative. Subtracting the dense equation from (72), and applying (79) and (80), gives on the stopped interval

$$\sup_{s \leq t} d_n(\widehat{\theta}(s), \theta_D(s)) \leq C_* e^{a_T s_{n,T}} \left[B_0 q^{-3/2} + s_{n,T} q^{-2} \right]. \quad (81)$$

Indeed the mobility error is bounded by its nondecreasing accumulated defect plus $C_T s_{n,T}$ times its time integral; replacing the defect by its terminal value and applying an integrating factor proves this form of Gronwall's inequality. Multiplication by $\sqrt{L+1}$ gives the stated bound for d_n , absorbed in C_* . The constants C_*, a_T are independent of n, q .

Choose $C_T = C_* \geq 1$ and $\Lambda_T \geq \max\{1, a_T, \log(4C_*)\}$ in the theorem. Then $H_{n,T} \geq s_{n,T}$ and $H_{n,T} \geq 4C_*$. Under (64), multiplying the right side of (81) by q bounds each term by C_* , and $q \geq s_{n,T} H_{n,T} \geq 4C_*$. Hence

$$\sup_{s \leq t} d_n(\widehat{\theta}(s), \theta_D(s)) \leq 2C_*/q \leq 1/2.$$

Since $|\widehat{\theta} - \theta_D|_n \leq d_n(\widehat{\theta}, \theta_D)$, continuity rules out a first exit at mobility distance one. A finite maximal raw endpoint before T is also impossible: the physical parameters stay within a bounded radius of the dense path at each fixed width, $1 \leq \tau \leq A_c$, and every raw moment is the integral of its bounded finite-width history against $|p_k| \leq 1$. Thus the full finite raw state stays in a compact subset of $\tau > 0$. Its locally Lipschitz field is bounded there, giving a limiting state and local continuation at a putative endpoint. This proves (65) on all of $[0, T]$. Finally, $H_{n,T} \geq s_{n,T}$ and $B_0 \leq 1$ show that the simpler order conditions stated after (65) imply (64). $\square$

Corollary B.3 (Predictions). *Under Proposition B.1, training predictions have the same C_T/q bound. If additionally $\|W_0^{(1)}\|_F/\sqrt{n} \leq A_1$, then for any probability measure μ on inputs with finite second moment,*

$$\left[\int \sup_{t \leq T} |\hat{f}_{n,q}(t, x) - f_{n,D}(t, x)|^2 d\mu(x) \right]^{1/2} \leq \frac{C_{T,A_1}}{q} \left[\int (1 + \|x\|_2/\sqrt{d})^2 d\mu(x) \right]^{1/2}.$$

The same estimate holds uniformly on any fixed bounded test-input set, with its radius replacing the moment factor.

Proof. Training predictions use the differential bound already proved on (74). The additional first-layer bound, dense energy and the distance bound control $\|W^{(1)}\|_F/\sqrt{n}$ on both paths and their joining segments. Linear activation growth and the bounded hidden operators imply feature RMS at most $C_{T,A_1}(1 + \|x\|_2/\sqrt{d})$ on those segments. Forward differentiation using bounded slopes, followed by the readout pairing, therefore gives $|Df(x)[v]| \leq C_{T,A_1}(1 + \|x\|_2/\sqrt{d})|v|_n$. Integrate along the parameter segment, apply (65), take the pointwise supremum in time, and then take the $L^2(\mu)$ norm. $\square$

Relation to the fixed-width guarantee. The fixed-width learning-speed theorem allows arbitrary $C_{\text{loc}}^{1,1}$ activations and constants depending on the particular width and initialization. The present result supplements it by assuming bounded slopes and uniform initial scales, and displaying which width dependence can be absorbed into the order. For unbounded activations the present proof establishes closure continuation only in (64); outside that region (81) is only a stopped estimate. No all-time estimate, population existence theorem, or width sampling rate follows from this finite-horizon argument.

C Proof of Theorem C.1

Theorem C.1 (Learning-speed clock). *Fix finite width, depth $L \geq 2$, arbitrary finite data and a finite initialization, and let every activation be in $C_{\text{loc}}^{1,1}(\mathbb{R})$. For every $T < \infty$ and all sufficiently large q the closure exists uniquely on $[0, T]$ and*

$$\sup_{0 \leq t \leq T} \|\hat{\theta}_q(t) - \theta(t)\| = O_T(q^{-1}),$$

where the threshold on q and the implied constant depend on T , the width, depth, data and initialization but not on q . The rate extends to hidden responses and to predictions on any bounded set of unseen inputs. If ϕ and ϕ' are bounded (tanh, sigmoid, arctangent, sine), it holds for every $q \geq 1$. No assumption of successful fitting, Gaussian initialization, input geometry or label structure is made.

In the fixed-width proofs we write $b_a^{(\ell)} = r_a \delta_a^{(\ell)}/\rho$ for the residual-weighted backward response, $h_a^{(\ell-1)}(\xi)$ and $b_a^{(\ell)}(\xi)$ for the forward and backward histories as functions of the clock coordinate ξ (extended over the unit prefix as in the main text), $\bar{h}_a^{(\ell-1)}$ and $\bar{\delta}_a^{(\ell)}$ for their moment matrices as in the main text, and $h_a^{(\ell-1)*}$, $b_a^{(\ell)*}$ for the values at the current endpoint of the degree- $(q-1)$ Legendre projections of the histories. The only approximation-theoretic input is the following Legendre tail bound.

Lemma C.2 (Legendre tail). *For $u \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|u - \Pi_q u\|_{L^2(0, \tau)}^2 \leq \frac{1}{q(q+1)} \int_0^\tau \xi(\tau - \xi) \|u'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4q(q+1)} \|u'\|_{L^2(0, \tau)}^2. \quad (82)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx}\{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Applying Bessel's inequality to u' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|u'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 u(x)p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq q(q+1)$ for $k \geq q$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (82); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\|\mathcal{D}^{-1/2}\dot{\theta}\right\|^2, \quad \int_0^t \left\|\dot{\theta}(s)\right\|^2 ds \leq n\rho(0)^2. \quad (83)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence would extend the solution, so the dense flow is global.

Now fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path stays at distance at least one from $\partial\mathcal{B}$. We stop the closure at its first exit from $\mathcal{B}$; the rest of the proof uses constants on this explicit ball and eventually rules out that exit.

To make the constants concrete, set $X = \max_a \|x_a\|/\sqrt{d}$ and $\alpha_0 = X$, and choose recursively

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi(z)|, \\ \beta_L &= Rs_L, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < L). \end{aligned} \quad (84)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2\right)^{1/2}, & \varrho &= R\alpha_L/n + Y, & S &= T\varrho, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_L, \\ V &= \left(v_1^2 + \sum_{\ell=2}^L v_\ell^2 + v_w^2\right)^{1/2}. \end{aligned} \quad (85)$$

On the stopped interval, $\rho \leq \varrho$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant Λ_F on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any single forward or backward history u , let $D_u(t) = \|u - \Pi_q u\|_{L^2(0, \tau(t))}^2$, and let u^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy with (7), every basis-dilation term cancels, leaving

$$\dot{D}_u = \rho \|u - u^*\|^2, \quad D_u(t) = \int_0^t \rho(s) \|u(s) - u^*(s)\|^2 ds. \quad (86)$$

The jump at the end of the backward prefix is harmless, since only square integrability and growing integrals are used.

Next differentiate (8). The physical closure equation is exactly

$$\dot{\hat{\theta}}_q = F(\hat{\theta}_q) + E, \quad E_1 = E_w = 0, \quad (87)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top. \quad (88)$$

The two mixed projection terms vanish by orthogonality, so the defect is a pure product of errors. Consequently

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell,a}(t) D_{h,\ell,a}(t)}. \quad (89)$$

This inequality is the bridge from a same-history approximation bound to a trajectory theorem. Define also the proof-only accumulator

$$W_{\text{acc}}^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t \hat{r}_a \hat{\delta}_a^{(\ell)} (\hat{h}_a^{(\ell-1)})^\top ds, \quad R_\ell = \widehat{W}^{(\ell)} - W_{\text{acc}}^{(\ell)}.$$

Then $R_\ell(0) = 0$ and $\dot{R}_\ell = E_\ell$, so (89) controls the accumulated absolute velocity error, not only a signed endpoint integral.

Step 3: forward histories have a q -independent derivative bound. Before changing variables, note that a nonstationary learning-speed solution cannot first reach $\rho = 0$ at a finite regular state. Every raw velocity vanishes there, and local uniqueness applied backward from that equilibrium would force the preceding path to be stationary. Hence $\xi = \tau(t)$ is a valid coordinate on every nonstationary compact segment; the raw algorithm itself remains defined at $\rho = 0$ without division.

Write primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 d\xi.$$

The first layer satisfies $\left\| (h_a^{(1)})' \right\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)} (h_a^{(\ell-1)})' \right]. \quad (90)$$

The delicate term is the defect, because endpoint evaluation of a polynomial projection can amplify errors by a factor that grows with q . Endpoint evaluation of a degree- $(q-1)$ polynomial has

operator norm $q/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < q} (2k + 1) = q^2$. Projection contraction and $m^{-1} \sum_a \|b_a^{(\ell)}\|^2 \leq \beta_\ell^2$ then imply

$$\left(m^{-1} \sum_a \|b_a^{(\ell)*}\|^2\right)^{1/2} \leq q\beta_\ell, \quad \left(m^{-1} \sum_a \|b_a^{(\ell)} - b_a^{(\ell)*}\|^2\right)^{1/2} \leq (q+1)\beta_\ell.$$

Combining this with Lemma C.2, (86) and (88) yields

$$\int_0^t \rho \|E_\ell / \rho\|_F^2 ds \leq \frac{2\beta_\ell^2 A^2}{n^2} Z_{\ell-1}(t). \quad (91)$$

Here the endpoint factor $q+1$ cancels the q^{-1} projection factor. This cancellation is the step that prevents a loss of convergence order with depth.

Define finite, q -independent bounds recursively by

$$\begin{aligned} \bar{Z}_1 &= S(s_1 v_1 X)^2, \\ \bar{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \bar{Z}_{\ell-1} \right]. \end{aligned} \quad (92)$$

Integrating (90), and using $\|x + y + z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ together with (91), proves inductively that $Z_\ell \leq \bar{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b,\ell,a} \leq S\beta_\ell^2$, and Lemma C.2 gives

$$\frac{1}{m} \sum_a D_{h,\ell,a} \leq \frac{A^2 \bar{Z}_{\ell-1}}{4q(q+1)}.$$

Summing (89) over links, we obtain

$$\int_0^t \|E(s)\| ds \leq \frac{B_{\text{LS}}}{\sqrt{q(q+1)}}, \quad B_{\text{LS}} = \frac{A}{n} \sum_{\ell=2}^L \beta_\ell \sqrt{S \bar{Z}_{\ell-1}}. \quad (93)$$

It remains to account for feedback. Subtracting the dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{LS}}}{\sqrt{q(q+1)}} + \Lambda_F \int_0^t e(s) ds,$$

and Gronwall's inequality yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{LS}} e^{\Lambda_F T}}{\sqrt{q(q+1)}}. \quad (94)$$

Choose q_0 so that the right side is less than $1/2$. The closure then cannot reach $\partial \mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the closure exists through T . Lipschitz dependence of the network responses on θ gives the final prediction statement.

Globally bounded activations. Suppose now that every activation and its first derivative are globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi\|_\infty, \quad s_\ell = \|\phi'\|_\infty, \quad B_w = \sqrt{\|w(0)\|^2 + nY^2 T}.$$

For both the dense and the closure flows, the unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2.$$

Therefore $\|w\| \leq B_w$, $\rho \leq \varrho := B_w \alpha_L / n + Y$, $S = T\varrho$ and $\tau \leq A := 1 + S$. Start with $\beta_L = B_w s_L$ and proceed downward: for $\ell = L, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (9) gives $\left\| \widehat{W}^{(\ell)} \right\|_F \leq D_\ell$, and the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity, because the descent starts from the bounded readout. Finally,

$$\left\| W^{(1)}(t) \right\|_F \leq \left\| W^{(1)}(0) \right\|_F + 2SX\beta_1.$$

These q -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ place the order- q state in a compact set for every fixed q , so the closure continues through T for every $q \geq 1$. Repeating Steps 2–4 with these global constants gives (94) for every q . This completes the proof. $\square$

D The joint feature-learning clock: construction and proof of Theorem D.1

Theorem D.1 (Joint feature-learning clock). *Under the setting of Theorem C.1, assume the activations of layers $\ell \geq 2$ are in $C_{\text{loc}}^{2,1}$ and $\rho(0) > 0$. For every T and all sufficiently large q the closure exists uniquely on $[0, T]$, its clock stays bounded independently of q , and $\sup_{t \leq T} \left\| \widehat{\theta}_q(t) - \theta(t) \right\| = O_T(q^{-2})$.*

Construction. Let Ψ be the concatenation over $\ell \geq 2$ and a of the forward and residual-weighted backward responses $(h_a^{(\ell-1)}, b_a^{(\ell)})$, so that the clock (10) reads $\dot{\tau} = g = \rho + \left\| \dot{\Psi} \right\|_2$ and $\|d\Psi/d\tau\| \leq 1$. History is placed at $\xi = \tau(s)$ with mass $\rho(s) ds$, while the prefix $[0, 1]$ carries Lebesgue mass and constant initial responses; the resulting measure μ_t satisfies $d\mu_t \leq d\xi$. With $p = (p_0, \dots, p_{q-1})^\top$ evaluated at ξ/τ , $e = (1, \dots, 1)^\top$, and the triangular matrix $\mathbf{T}_{kk} = k$, $\mathbf{T}_{kj} = 2j + 1$ for $j < k$ (zero above the diagonal), the state consists of the moment matrices $\bar{h}_a^{(\ell-1)} = \int h_a^{(\ell-1)}(\xi) p^\top d\mu_t$, $\bar{\delta}_a^{(\ell)} = \int b_a^{(\ell)}(\xi) p^\top d\mu_t$ and one shared Gram matrix $G = \int pp^\top d\mu_t$, which evolve by

$$\begin{aligned} \widehat{W}^{(\ell)} &= W_0^{(\ell)} - \frac{2}{nm} \sum_a (\bar{\delta}_a^{(\ell)} G^{-1} (\bar{h}_a^{(\ell-1)})^\top - b_a^{(\ell)}(0) h_a^{(\ell-1)}(0)^\top), \\ \dot{\bar{h}}_a^{(\ell-1)} &= \rho h_a^{(\ell-1)} e^\top - (g/\tau) \bar{h}_a^{(\ell-1)} \mathbf{T}^\top, \quad \dot{\bar{\delta}}_a^{(\ell)} = r_a \delta_a^{(\ell)} e^\top - (g/\tau) \bar{\delta}_a^{(\ell)} \mathbf{T}^\top, \\ \dot{G} &= \rho ee^\top - (g/\tau)(\mathbf{T}G + G\mathbf{T}^\top), \end{aligned} \tag{95}$$

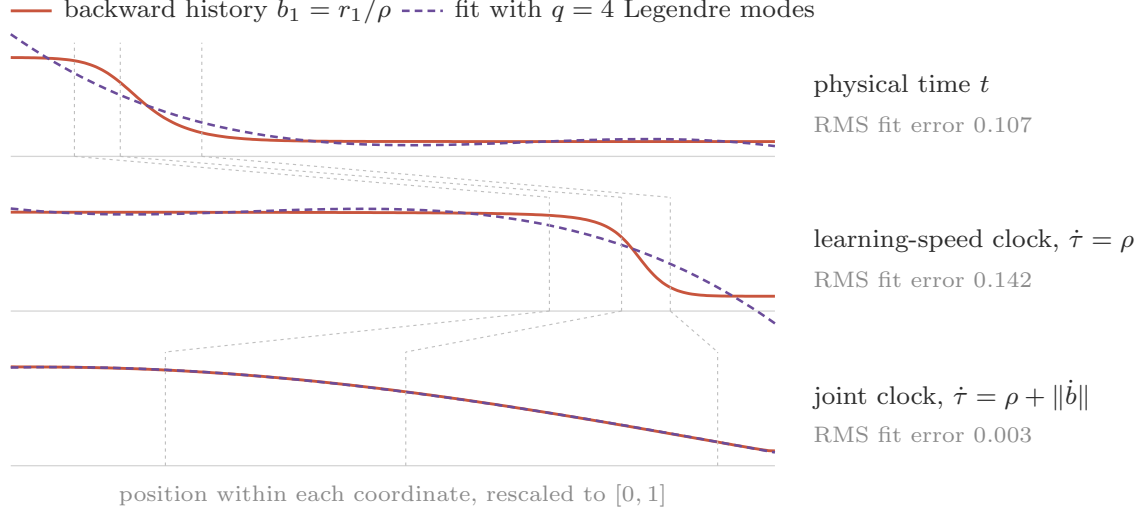


Figure 4: Why the clock matters, on a toy signal. Two residuals decay at different rates, $r_1 = e^{-4t}$ and $r_2 = 0.05 e^{-t/2}$, so the normalized backward signal $b = r/\rho$ switches direction late, when the loss is already small. Its first component is drawn in physical time, in the learning-speed clock and in the joint clock, each rescaled to $[0, 1]$, with a least-squares fit by four Legendre modes; dotted lines join the same physical instants. The learning-speed clock compresses the late phase and sharpens the switch; the joint clock spreads it out. The three fit errors are measured in three different coordinates, so they illustrate how smooth the history is in each, not a common-metric comparison of closure errors; the joint closure itself uses the residual-weighted projection with a Gram matrix.

with $G(0) = \text{diag}(1, \frac{1}{3}, \dots, \frac{1}{2q-1})$, only the zeroth moment columns nonzero initially, and the prefix product subtracted so that $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$. With $h_a^{(\ell-1)*} = \bar{h}_a^{(\ell-1)} G^{-1} e$ and $b_a^{(\ell)*} = \bar{\delta}_a^{(\ell)} G^{-1} e$, differentiating the reconstruction cancels every g/τ term and gives the defect $E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top$, of the same product form as (88).

Step 1: a lower bound on the dense residual. Assume $\rho_0 > 0$ as stated, and use the compact physical ball $\mathcal{B}$ from Appendix C. Retain the constants $\alpha_\ell, \beta_\ell, \varrho, S, \mathbf{A}, V, \Lambda_F$ from the proof of Theorem C.1. On this ball the output map has a finite Lipschitz constant c_f : prediction and residual RMS change by at most $c_f \|\theta - \theta'\|$. Along dense flow,

$$|\dot{\rho}_{\text{dense}}| \leq c_f \|F(\theta)\| \leq c_f V \rho_{\text{dense}}.$$

Consequently

$$\rho_{\text{dense}}(t) \geq \mu := \rho_0 e^{-c_f V T} > 0, \quad 0 \leq t \leq T. \quad (96)$$

Step 2: an exact weighted projection energy. Consider the maximal regular closure interval stopped before either its physical state leaves $\mathcal{B}$ or its residual reaches $\mu/2$. We first derive a clock bound on this stopped interval without assuming small projection error. For any inserted vector history f , let

$$D_f = \|f - \Pi_t f\|_{L^2(\mu_t)}^2 = \int \|f\|^2 d\mu_t - \text{tr}(B_f G^{-1} B_f^\top).$$

Differentiating $G^{-1}G = I$ gives

$$\frac{d}{dt} G^{-1} = -\rho G^{-1} e e^\top G^{-1} + (g/\tau)(G^{-1} \mathbb{T} + \mathbb{T}^\top G^{-1}).$$

Together with (95), this cancels every dilation term and yields

$$\dot{D}_f = \rho \|f - f^*\|^2, \quad D_f(t) = \int_0^t \rho \|f - f^*\|^2 \, ds. \quad (97)$$

The initial error is zero because every prefix is constant. Thus (89) holds with the weighted errors in (97).

Step 3: order-independent coarse bounds. Put

$$Q = m \sum_{\ell=2}^L (\alpha_{\ell-1}^2 + \beta_\ell^2), \quad C_0 = \frac{AQ}{nm}.$$

The total mass of μ_t is at most $A = 1 + S$. Projection error is no larger than raw history energy, including the matching prefixes. Applying $2\sqrt{uv} \leq u + v$ in the defect estimate gives the order-independent bounds

$$\int_0^t \|E\| \, ds \leq C_0, \quad \int_0^t \|\dot{\hat{\theta}}\| \, ds \leq VS + C_0. \quad (98)$$

The second inequality uses $\dot{\hat{\theta}} = F(\hat{\theta}) + E$ and $\int_0^t \|F\| \, ds \leq V \int_0^t \rho \, ds \leq VS$. This total-variation estimate is stronger than bounded physical position, and it is obtained before any clock bound.

Step 4: a clock bound without a cap. It remains to control the derivative of the normalized response map. Let $\tilde{r} = r/\rho \in \mathbb{R}^m$. Then

$$\|\tilde{r}\|^2 = m, \quad D\tilde{r}[v] = \left(I - \frac{\tilde{r}\tilde{r}^\top}{m} \right) \frac{Dr[v]}{\rho}. \quad (99)$$

The matrix in parentheses is an orthogonal projector, and $\rho \geq \mu/2$ on the stopped interval. Differentiating the forward and backward recurrences uses only bounded physical weights and bounded first two activation derivatives on compact preactivation intervals; the first layer uses only ϕ' . Hence there is a finite, q -independent constant J such that

$$\|D\Psi(\vartheta)\|_{\text{op}} \leq J$$

throughout the stopped tube. The stated local smoothness also makes this derivative locally Lipschitz. Since $\dot{\Psi} = D\Psi(\hat{\theta})\hat{\dot{\theta}}$, (98) now gives

$$\tau_q(t) = 1 + \int_0^t (\rho + \|\dot{\Psi}\|) \, ds \leq A + J(VS + C_0) =: \Lambda_T. \quad (100)$$

No clock cap has been assumed or inserted.

Step 5: the sharp joint approximation estimate. We can now use the sharper approximation estimate. The entire concatenated history $\overline{\Psi}(\xi)$ is constant on the prefix, continuous at its join and, on the physical history,

$$\|\partial_\xi \overline{\Psi}\| = \frac{\|\dot{\Psi}\|}{\rho + \|\dot{\Psi}\|} \leq 1.$$

Therefore $\int_0^{\tau_q} \|\overline{\Psi}'\|^2 d\xi \leq \tau_q - 1$. The weighted projection is a best approximation in $L^2(\mu_t)$, and $d\mu_t \leq d\xi$, so comparison with the ordinary Legendre projection and Lemma C.2 give the joint bound

$$\sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{\tau_q^2(\tau_q - 1)}{4q(q+1)} \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4q(q+1)}. \quad (101)$$

This is one estimate for the concatenated path; it does not introduce a separate dimension factor per unit or layer.

Step 6: feedback and continuation. Combining (89), (100), (101) and $2\sqrt{uv} \leq u + v$ yields

$$\int_0^t \|E\| ds \leq \frac{\Gamma_{\text{JF}}}{q(q+1)}, \quad \Gamma_{\text{JF}} = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (102)$$

Subtracting the dense flow and applying the Lipschitz constant Λ_F and Gronwall's inequality gives

$$\|\hat{\theta}_q(t) - \theta(t)\| \leq \frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)}. \quad (103)$$

Choose $q_0(T)$ so that, for every $q \geq q_0(T)$,

$$\frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)} \leq \frac{\mu}{4}. \quad (104)$$

The first inequality rules out physical exit, because the dense path stays at distance one from $\partial\mathcal{B}$. The second, together with (96), gives $\hat{\rho} \geq 3\mu/4 > \mu/2$, ruling out residual exit.

Finally, the matching prefix supplies, at every fixed q ,

$$G(t) \succeq \int_0^1 p(\xi/\tau_q(t)) p(\xi/\tau_q(t))^\top d\xi. \quad (105)$$

The right side is positive definite, since a zero quadratic form would make a nonzero polynomial vanish on an interval. Its entries vary continuously with τ_q , so on $1 \leq \tau_q \leq \Lambda_T$ its least eigenvalue has a positive minimum $\gamma_{q,T}$; this minimum need not be uniform in q . The moment integral representations, finite history mass, bounded responses and bounded fixed-order polynomials then bound every raw coordinate. Together with the physical, residual, clock and Gram bounds, the full state lies in a compact subset of its locally Lipschitz domain. A finite maximal endpoint therefore extends, which proves continuation through T and completes the theorem, with implied constant $\Gamma_{\text{JF}} e^{\Lambda_F T}$. $\square$

Implementation of the joint clock

Equation (10) appears implicit only because $\dot{\Psi}$ is written compactly; no nonlinear solve for g is needed. The internal velocities do not depend on g , so they can be computed first. Begin with $G^{-1}e$, the endpoint fits h^*, b^* , and the internal velocities

$$V_\ell = \hat{W}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_a^{(\ell)} (h_a^{(\ell-1)*})^\top + b_a^{(\ell)*} (h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top \right]. \quad (106)$$

These expressions contain no g . Next compute $V_1 = F_1$ and $V_w = F_w$, and run a directional forward pass,

$$\dot{z}_a^{(1)} = V_1 x_a / \sqrt{d}, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \hat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)},$$

which gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass then differentiates the backward recurrence, using ϕ'' for $\ell \geq 2$, and yields

$$\dot{b}_a^{(\ell)} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_a^{(\ell)} \frac{\dot{\rho}}{\rho}.$$

Only then do we evaluate $g = \rho + \|\dot{\Psi}\|$ and, finally, (95). Matrix and transpose actions use the same reconstructed operator. In practice one factors G and solves linear systems; an explicit inverse is unnecessary.

The joint clock is more regular in approximation coordinates, but it is not automatically better at low order. At $q = 1$ the polynomial space consists of constants, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

E Exact compression in a deep linear benchmark

Take one datum $x = 1$, a label y , three linear hidden layers with order-one initialization, mobilities $(n\kappa, \kappa, \kappa, n\kappa)$, and the normalized factors $u = W^{(1)}/\sqrt{n}$, $v = W^{(4)}/\sqrt{n}$, so that $f_n = v^\top W^{(3)} W^{(2)} u$. Packing the four factors into the cyclic block matrix $\mathcal{C}_n$ with blocks $v^\top, u, W^{(2)}, W^{(3)}$ below and above the diagonal gives $f_n = \frac{1}{4} \text{Tr} \mathcal{C}_n^4$, and the feature derivative of $\mathcal{C}_n$ is $(\mathcal{C}_n^\top)^3$.

Theorem E.1 (Exact operator closure). *There are a separable Hilbert space, a bounded initialized source $\mathcal{C}_0$ and a unique global trace-class increment $Q(t)$ with*

$$\mathcal{C} = \mathcal{C}_0 + Q, \quad f = \frac{1}{4} \text{Tr} \mathcal{C}^4, \quad \dot{Q} = -2\kappa(f - y)(\mathcal{C}_0^* + Q^*)^3, \quad Q(0) = 0.$$

The kernel is $K = \text{Tr}[\mathcal{C}^3(\mathcal{C}^)^3]$, the residual obeys $\dot{r} = -2\kappa r K$, and $\|Q(t)\|_1 \leq 2|y|\sqrt{\kappa t}$. For every fixed T , finite gradient flow, and gradient descent for every step sequence tending to zero, converge in probability to $(f, K, \mathcal{L})$ uniformly on $[0, T]$.*

The proof builds $\mathcal{C}_0$ as the limit of initialized word statistics on a colored Fock space, obtains global existence from trace-norm control, passes to the width limit through finite-word Picard iterates, and treats gradient descent as an exact Euler step for Q .

Theorem E.2 (Restricted scalar nonclosure). *In the same model, no encoder formed from a fixed finite list of complete tensor contractions of $(u, W^{(2)}, W^{(3)}, v)$ with connected-graph degree bounded independently of width admits a width-independent polynomial autonomous ODE (or finite-order local polynomial PDE) and a polynomial output readout reproducing the output dynamics on a nonempty open set of states for all sufficiently large widths.*

The $(2j - 1)$ st feature derivative of the output contains the connected contraction

$$u^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j u$$

of degree $4j + 2$. Polynomial operations on a bounded-degree alphabet form only disjoint unions of connected components already present, and nonisomorphic contraction graphs are linearly independent at large width, so this term cannot be synthesized. The theorem does not exclude arbitrary encoders, unrestricted fields, or closures tailored to a single initialized trajectory.

F Accuracy and resource comparisons: derivations

All comparisons concern the same dense population predictor f_∞ , with the canonical mobility, data, Gaussian initialization and exactly zero readout of Theorem 3.1. Matching initial predictions alone would not ensure this agreement. Moving state means persistent updated scalars; fixed initialized matrices, coefficient evaluators and temporary workspace are accounted for separately. Constants may depend on the fixed task, including m, d, L , the Gram margin, activation bounds and test law.

Definition F.1 (Size at accuracy). For a specified representation and a stated sufficient error certificate, its size at accuracy ε is the least moving state obtained by choosing the representation's resolution parameters to satisfy that certificate, at the specified horizon and confidence level. This is a sufficient representation cost, not an algorithmic lower bound.

F.1 The all-time certificate and its quantitative limits

For $0 < T \leq \infty$, use the finite-horizon restriction of (11),

$$E_\mu(g, f; T) = \left[\int \sup_{0 \leq t \leq T} |g(t, x) - f(t, x)|^2 \mu(\mathrm{d}x) \right]^{1/2}. \quad (107)$$

The test law has finite second input moment. This whole-input metric controls the difference of test RMSEs against any target in $L^2(\mu)$, uniformly in time; it does not assert that f_∞ generalizes well. We use $T = \infty$ for the response-memory theorem. Finite- T certificates for other methods below remain finite- T comparisons unless an additional tail estimate extends them to this same all-time metric.

Let $D_n = E_\mu(f_n, f_\infty; \infty)$. Prediction transfer from the normalized distance (3), followed by the triangle inequality, gives on the common initialization events

$$E_\mu(\hat{f}_{n,q}, f_\infty; \infty) \leq C_\mu \omega(q) + B_{n,\mu} + D_n, \quad \omega(q) = q^{-2} \exp\left\{K \sqrt{\log(e+q)}\right\}. \quad (108)$$

Here $B_{n,\mu}$ is the closure's dense-carrier transfer remainder after prediction transfer. Both $B_{n,\mu}$ and D_n tend to zero in probability; the theorem supplies no numerical width rate for either. Their sum is the order-independent remainder $b_{n,\mu}$ in Theorem 3.1. The events and remainders are common to all q , so every deterministic $q_n \rightarrow \infty$ gives convergence in the all-time metric without an interchange of fixed-order limits.

The moving-state counts are

$$M_{\text{dense}}(n) = (L-1)n^2 + n(d+1), \quad M_{\text{mem}}(n, q) = 2(L-1)mnq + n(d+1) + O(1). \quad (109)$$

For fixed positive integers m, d and fixed $L \geq 2$, $q_n \rightarrow \infty$ and $q_n = o(n)$ imply $M_{\text{mem}}/M_{\text{dense}} \rightarrow 0$, proving Corollary 3.2. For example, $q_n = \lceil \log \log(e^e + n) \rceil$ already suffices. This is a convergence and compression statement along the same width sequence, not a comparison of the narrowest networks attaining a tolerance.

Conditional costs at a fixed confidence. An additional quantitative benchmark is

$$D_n = O_{\text{Pr}}(n^{-1/2}), \quad B_{n,\mu} = O_{\text{Pr}}(n^{-1/2}). \quad (110)$$

The second hypothesis does not follow from the first: controlling dense predictions need not quantify the empirical transfer of dense backward carriers. Neither follows merely from a fixed-program

population limit or the perturbative fluctuation calculation of Bordelon and Pehlevan (2023). These hypotheses are needed in the same all-time metric and probability convention. In particular, convergence in probability in Theorem 3.1 is not an unconditional RMS rate.

Under (110), at any fixed confidence one may take $n = O(\varepsilon^{-2})$ with a sufficiently large constant and choose q so that $C_\mu \omega(q) \leq \varepsilon/2$. Writing $z = \log q$ gives $\log \omega(q) = -2z + K\sqrt{z} + o(1)$, hence a sufficient order satisfies

$$\log q = \frac{1}{2} \log(1/\varepsilon) + O\left(\sqrt{\log(1/\varepsilon)}\right), \quad q = \varepsilon^{-1/2+o(1)}. \quad (111)$$

The constant in the $O(\sqrt{\log(1/\varepsilon)})$ correction is chosen sufficiently large to enforce the error inequality. Consequently, for fixed m, d, L ,

$$M_{\text{mem}} = O\left(\varepsilon^{-5/2+o(1)}\right), \quad M_{\text{dense}} = O(\varepsilon^{-4}). \quad (112)$$

These are sufficient costs from the two certificates; no optimality or lower bound is asserted. Without (110), Corollary 3.2 still holds, but (112) is not a proved accuracy-cost law.

For reference, a pure-power certificate $An^{-1/2} + Bq^{-p} \leq \varepsilon$, $p > 0$, leads to the continuous allocation problem $\min u^{-2}(\varepsilon - u)^{-1/p}$, where $u = A/\sqrt{n}$. Its logarithmic derivative is $-2/u + 1/[p(\varepsilon - u)]$, with positive derivative, and its objective diverges at both endpoints. Thus

$$u_* = \frac{2p\varepsilon}{2p+1}, \quad n_* = \frac{(2p+1)^2 A^2}{4p^2 \varepsilon^2}, \quad q_* = \left(\frac{(2p+1)B}{\varepsilon}\right)^{1/p}. \quad (113)$$

Upward rounding preserves feasibility. The near-quadratic theorem has a subpower correction and an unquantified width remainder, so the exact $p = 2$ allocation is not its unconditional optimum.

The original residual-speed implementation permits full-batch arithmetic per right-hand-side evaluation

$$C_{\text{dense}} = O(Lmn^2 + mnd), \quad C_{\text{mem}} = O(Lmn^2 + Lm^2 nq + Lmnq + mnd). \quad (114)$$

The terms account for initialized matrix actions, learned-factor actions, moment updates and first-layer work. The closure also retains $(L-1)n^2$ initialized hidden entries as fixed data; activation workspace is $O(Lmn)$. The theorem therefore compresses evolving state, without an automatic improvement of total storage or arithmetic at a prescribed accuracy.

F.2 A matched tangent hierarchy

For sufficiently differentiable activations, the canonical mobility $\mathcal{D}_n = \text{diag}(nI, I, \dots, I, nI)$ defines

$$K^{(2)}(x, z) = \nabla_\theta f(x)^\top \mathcal{D}_n \nabla_\theta f(z), \\ K^{(j+1)}(x_1, \dots, x_{j+1}) = (\nabla_\theta K^{(j)}(x_1, \dots, x_j))^\top \mathcal{D}_n \nabla_\theta f(x_{j+1}).$$

The chain rule gives the exact matched hierarchy

$$\dot{f}(x) = -\frac{2}{m} \sum_a r_a K^{(2)}(x, x_a), \quad \dot{K}^{(j)} = -\frac{2}{m} \sum_a r_a K^{(j+1)}(\dots, x_a). \quad (115)$$

Truncation through order $q \geq 2$ retains exact initial coefficients and freezes $K^{(q)}$. A population implementation additionally needs limiting initial coefficients and a valid limiting hierarchy for this mobility. The $C^{1,1}$ activation class of Theorem 3.1 alone does not supply the higher derivatives needed to define arbitrary orders.

A conditional finite-time certificate. Suppose on $[0, T]$ that $|K^{(j)}| \leq B_j$ when the first argument ranges over the training inputs and a controlled test set, and all other arguments are training inputs. Set $R = 1 + \sup_{t,a} |r_a(t)|$ and stop the truncated flow at predictor error one. Its residuals are then bounded by R . For the corresponding supremum predictor and kernel errors,

$$\begin{aligned} e_q(t) &\leq 2RB_{q+1}t, \\ e_j(t) &\leq \int_0^t (2Re_{j+1} + 2B_{j+1}e_f) ds \quad (2 \leq j < q), \\ e_f(t) &\leq \int_0^t (2Re_2 + 2B_2e_f) ds. \end{aligned} \tag{116}$$

Successive substitution integrates the top forcing q times. The feedback kernels are

$$2B_{k+2} \frac{(2R(t-s))^k}{k!}, \quad 0 \leq k \leq q-2.$$

Bounding these at T and applying Gronwall yields

$$\sup_{t \leq T} e_f(t) \leq B_{q+1} \frac{(2RT)^q}{q!} \exp(\Lambda_{q,T}T), \quad \Lambda_{q,T} = 2 \sum_{k=0}^{q-2} B_{k+2} \frac{(2RT)^k}{k!}. \tag{117}$$

If this bound is below one, it rules out first exit, while the triangular integral equations bound all training tensors and ensure continuation. It controls (107) for a test law supported on the controlled set; an unbounded test law needs an integrable envelope. Random coefficients and their approximation errors need the same probability convention as the compared predictor.

For example, the additional factorial-growth and stability conditions $B_j \leq UV^{j-2}(j-2)!$ and $\chi = 2RVT < 1$ give $\Lambda_{q,T} \leq 2U/(1-\chi)$ and error at most $D\chi^q/q$, where $D = (U/V) \exp\{2UT/(1-\chi)\}$. For this certificate, with $\lambda_\chi = \log(1/\chi)$,

$$q_\varepsilon = \max \left\{ 2, \left\lceil \frac{W(\lambda_\chi D/\varepsilon)}{\lambda_\chi} \right\rceil \right\}, \quad M_{\text{NTH}} = \sum_{j=1}^{q_\varepsilon-1} m^j = \frac{m^{q_\varepsilon} - m}{m-1} \quad (m \geq 2). \tag{118}$$

Here $W(z)e^{W(z)} = z$. The fixed top tensor and direct work each have order m^{q_ε} . At fixed $m \geq 2$ and χ the moving count is of order $[D/(\varepsilon \log(D/\varepsilon))]^{\log m/\lambda_\chi}$. The exponent depends on a certified χ ; a universal $\log_2 m$ exponent does not follow. Analyticity of $\tanh$ by itself does not establish these bounds, and $\chi \geq 1$ makes this particular certificate inconclusive. Population use needs resource-independent constants and access to the initial mixed test-input kernels. No all-time conclusion follows from this finite- T argument.

What the published NTH theorem supplies. In their native NTK parameterization, Huang and Yau (2020, Theorem 2.6) prove, after exchanging width/sample symbols,

$$\begin{aligned} \frac{\|f(t) - \tilde{f}(t)\|_2}{\sqrt{m}} &\lesssim \frac{(1+t)t^{q-1}}{n^{q/2}} \min\{t, m/\lambda\}, \\ t &\leq \min \left\{ \frac{c\sqrt{\lambda n/m}}{(\log n)^{C'}}, \frac{n^{q/[2(q+1)]}}{(\log n)^{C'}} \right\}. \end{aligned} \tag{119}$$

This assumes their initialization, even q , bounded activation derivatives through $2q + 1$, bounded nondegenerate inputs with uniformly independent subsets through size $2q + 1$, and an initial kernel eigenvalue gap. Loss normalization only rescales time by a constant. The estimate concerns training RMS in that parameterization; it supplies neither a whole-input all-time bound for the canonical feature-learning target nor the factorial and stability hypotheses above.

For whole-function queries the matched hierarchy has a finite coefficient representation. Set $u_a = -2\tilde{r}_a/m$, $I_\emptyset = 1$ and $\dot{I}_{a_1 \dots a_k} = u_{a_1} I_{a_2 \dots a_k}$, with zero initial values for nonempty words. Repeated integration gives

$$\tilde{f}_q(t, x) = f_0(x) + \sum_{k=1}^{q-1} \sum_{a_1, \dots, a_k} K_0^{(k+1)}(x, x_{a_1}, \dots, x_{a_k}) I_{a_1 \dots a_k}(t). \quad (120)$$

The moving count agrees with (118); query work additionally includes the fixed mixed-kernel evaluators. For $m = 1$, all coefficients are $I_k = s^k/k!$ with $\dot{s} = u$, so one moving scalar suffices. Thus the direct tensor count is a representation cost, not a universal lower bound.

F.3 A specified streamed DMFT solver and stored histories

The self-consistency solver of Bordelon and Pehlevan (2022, Appendix B, Algorithm 1) stores covariance and response kernels indexed by input and time pairs. Matching scales requires $\gamma = \sqrt{n}$, internal coordinates $U^{(\ell)} = \sqrt{n}W^{(\ell)}$, and the same readout law and loss normalization. A fixed-program Tensor Programs limit (Yang and Hu, 2021, Algorithm 1, Theorem 7.4) does not itself prescribe a convergent numerical solver or quantify its error.

For N_t time steps, S sampled paths and I self-consistency sweeps, factoring a dense covariance costs $O((mN_t)^3)$ per layer. Streaming one path and its full response Jacobian while accumulating kernel averages gives the implementable upper bounds

$$M_D = O(Lm^2N_t^2), \quad C_{D,\text{total}} = O(IL(S+1)m^3N_t^3), \quad C_{D,\text{step}} = O(IL(S+1)m^3N_t^2). \quad (121)$$

Memory includes kernel and Jacobian workspace; the last cost is amortized. Mixed kernels and test paths must also be available for test prediction. These are costs for this solver, not lower bounds for TP or DMFT.

A sufficient stability certificate would be a discrete self-consistency map $\mathcal{F}$ that is a κ -contraction, $\kappa < 1$, in a norm controlling test outputs. If its sampled approximation differs uniformly by ζ and a computed point has residual at most η_0 , then

$$\|\hat{z} - z_*\| \leq \eta_0 + \zeta + \kappa \|\hat{z} - z_*\| \implies \|\hat{z} - z_*\| \leq \frac{\eta_0 + \zeta}{1 - \kappa}.$$

With a Lipschitz output map, uniform sampling control and a stable order- σ time discretization, this would yield, at the specified confidence and finite horizon,

$$E_\mu(g_{S,N_t}, f_\infty; T) \leq A_D S^{-1/2} + B_D (T/N_t)^\sigma + \eta. \quad (122)$$

The constants include stability amplification. Grid-uniform Monte Carlo control, a certified numerical order, self-consistency stability and whole-input output control are additional hypotheses, not consequences of the cited limiting descriptions. An all-time comparison further requires a certified treatment of the remaining time tail.

Conditionally on (122) for a fixed σ , assigning fixed fractions of ε to the three terms gives $S = O(\varepsilon^{-2})$, $N_t = O(T\varepsilon^{-1/\sigma})$ and $M_D = O(Lm^2T^2\varepsilon^{-2/\sigma})$. Streaming moves the sampling cost into

work; it does not eliminate it. One cannot improve this exponent by declaring arbitrarily high numerical order without the corresponding regularity, stability, constants and stage costs. There is no certified universal DMFT accuracy exponent here.

A separate finite-network baseline stores every Euler update:

$$W_k^{(\ell)} = W_0^{(\ell)} - \frac{2\Delta}{nm} \sum_{j < k, a} r_{a,j} \delta_{a,j}^{(\ell)} (h_{a,j}^{(\ell-1)})^\top.$$

Its moving memory is $O(LmnN_t)$, fixed memory $O(Ln^2)$ and late-step work $O(Lmn^2 + Lm^2nN_t)$. A specified method with a fixed number of stages has the same storage order. If it also has a matched certificate $A/\sqrt{n} + D_h(T/N_t)^\sigma$, (113) gives sufficient moving state $O(LmA^2TD_h^{1/\sigma}\varepsilon^{-(2+1/\sigma)})$. This conditional finite- T calculation stores finite-network updates; it is distinct from sampling a population response system. Its numerical order and horizon dependence need their own certificates.

The resource formulas identify what each representation stores and which additional estimates would make accuracy comparisons quantitative. Only Theorem 3.1 and corollary 3.2 supply the all-time response-memory convergence claim used here; none of the conditional exponents establishes a universal ordering or an optimal complexity theorem.

G Supplementary fitted functions on the sphere

Figure 5 adds a three-dimensional-input, four-hidden-layer example. The sphere RMS discrepancies from dense are 0.02193, 0.01280, and 0.00454 for $q = 1, 2, 3$. Dense and the three closures reach training RMS ≤ 0.01 at times 60, 75.5, 63.25, and 60, respectively. This is supplementary fitted-function evidence outside the smooth-activation theorems: ReLU is nonsmooth, and the endpoints are individually fitted. Rendering the saved predictions does not independently reproduce training.

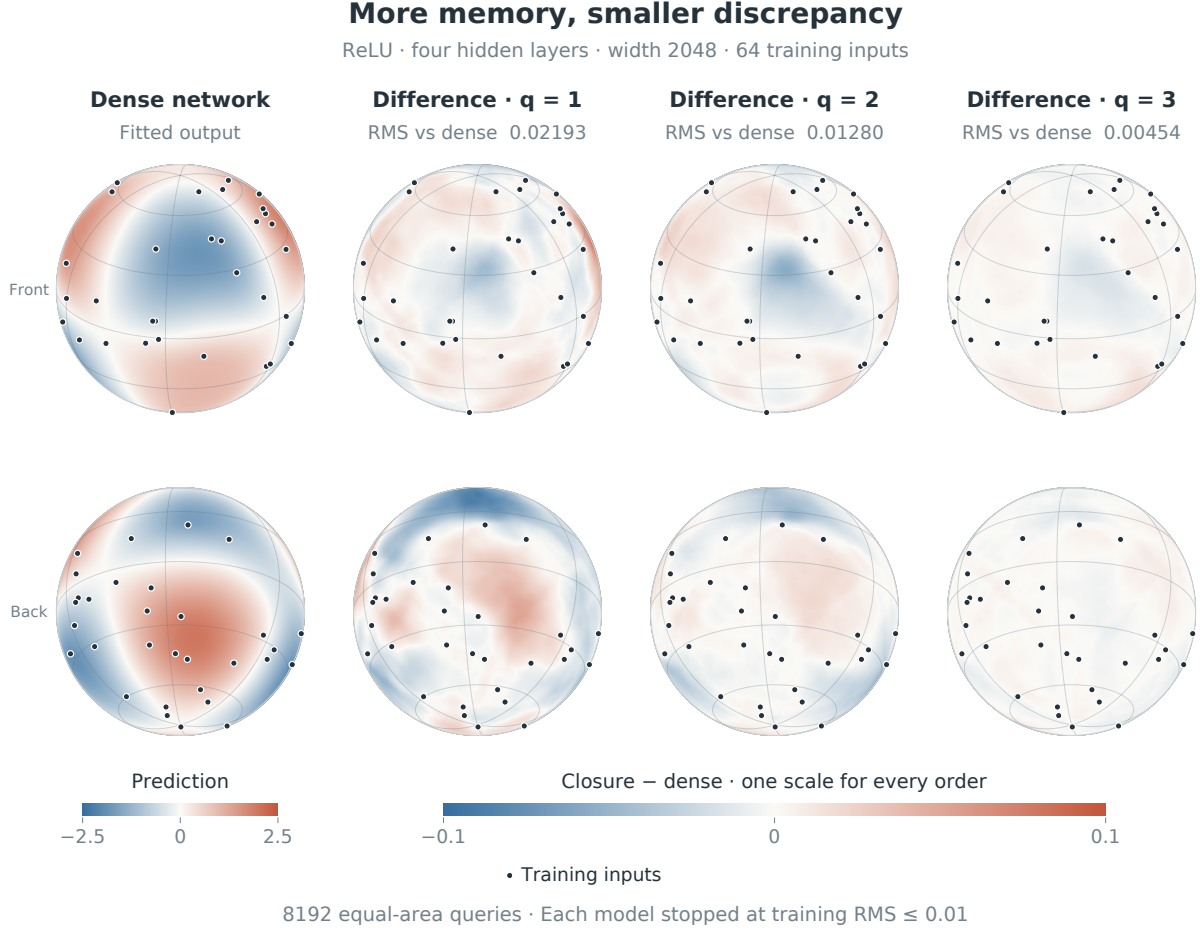


Figure 5: Saved fitted functions on the unit sphere for the target $\sqrt{105}x_1x_2x_3$, with four hidden ReLU layers of width 2048 and 64 training inputs. The first column shows the dense output; the remaining columns show signed closure-minus-dense differences for $q = 1, 2, 3$. Rows show complementary hemispheres, and black dots mark training inputs. Output colors use $[-2.5, 2.5]$; every error panel uses $[-0.1, 0.1]$. Surface colors interpolate saved queries on a triangular mesh; RMS values use the original 8192 equal-area query values before interpolation. Models are compared at their individual training-RMS ≤ 0.01 endpoints, using one network seed and float32 Euler steps of $1/128$. The order trend is specific to this one-seed comparison. These are fitted-function discrepancies, without a common-time trajectory comparison or a continuous-gradient-flow refinement certificate.

Table 2: Whole-circle RMS difference between response memory and dense training at individually fitted endpoints. Left: two hidden tanh layers, width 2048 (Figure 6). Right: three hidden tanh layers, width 4096 (Figure 2).

Task	two layers			three layers		
	$q = 1$	$q = 3$	$q = 7$	$q = 1$	$q = 2$	$q = 3$
Two outliers, alternating	0.236	0.0730	0.00466	0.108	0.0238	0.0315
Quadrant, alternating	0.341	0.0456	0.00155	0.275	0.0284	0.0585
Quadrant, paired	0.0193	0.00282	0.00012	0.0122	0.00371	0.00121
Quadrant, center/edges	0.0456	0.00488	0.00131	0.0790	0.0257	0.00725
Equally spaced, mixed	0.0124	0.00053	4×10^{-6}	0.0150	0.00241	0.00028

H Supplementary two-hidden-layer circle gallery

Figure 6 shows the full two-hidden-layer task gallery, and Table 2 collects the whole-circle RMS discrepancies of both depths (the three-hidden-layer gallery is Figure 2 in the main text).

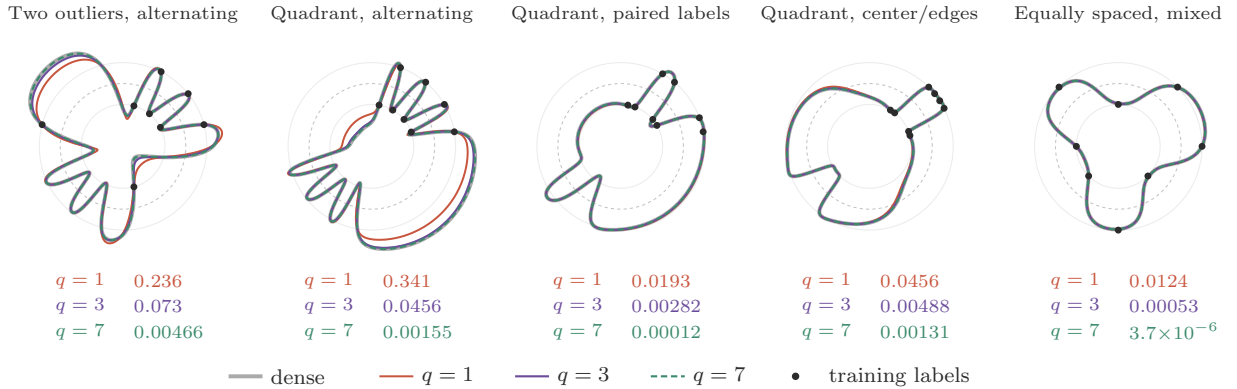


Figure 6: Fitted circle predictions for two hidden tanh layers, width 2048, and response-memory orders $q = 1, 3, 7$, using the dense references in Table 2. Models are evaluated at their individual training-MSE 10^{-3} endpoints. Angle is input direction and radius is $3 + f(\theta)$; rings, training labels and RMS values follow Figure 2. RMS is measured on 8192 uniform circle points.

I Supplementary experimental figures

Figure 7 sketches the paired memory and the product form of its defect; Figure 8 shows what Legendre moments retain of recorded response histories; Figure 9 shows the RMS discrepancies behind Figure 1 at every shared time; Figure 10 shows the three-hidden-layer order trends of Table 2 next to their numerical sensitivity and the movement of the hidden features; Figure 11 shows the MNIST predictions and residuals summarized in Section 4.

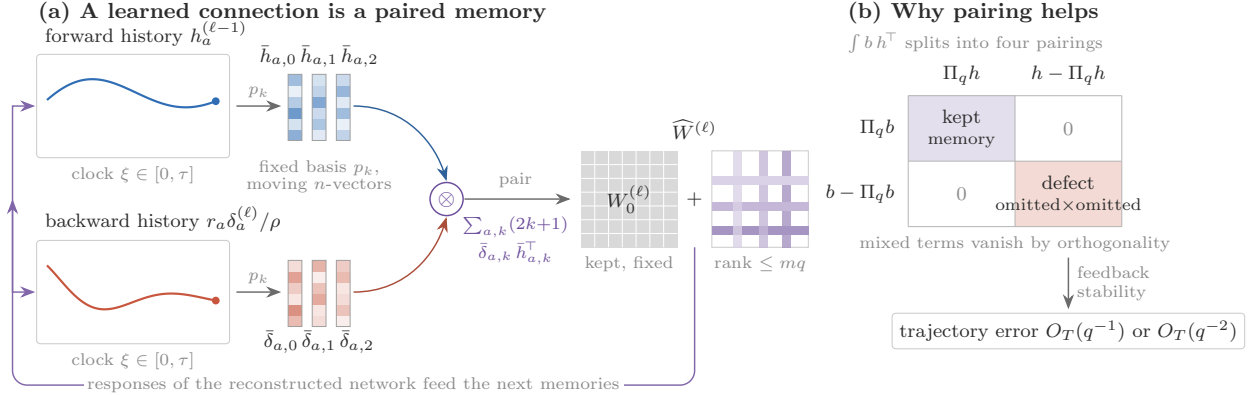


Figure 7: A learned connection is a paired memory. (a) For each hidden link and training sample, the forward response $h_a^{(\ell-1)}$ and the residual-weighted backward response $b_a = r_a \delta_a^{(\ell)} / \rho$ are recorded in the learning clock and projected onto the first q shifted Legendre polynomials ($q = 3$ shown). The polynomial coordinates are fixed; the moment vectors $\bar{h}_{a,k}, \bar{\delta}_{a,k} \in \mathbb{R}^n$ keep moving. Pairing them gives a correction of rank at most mq , added to the fixed initialization $W_0^{(\ell)}$, and the responses of the reconstructed network feed the next memories. (b) Splitting each history into its projection Π_q and remainder, the exact interaction $\int b h^\top$ is a sum of four pairings; the mixed ones vanish by orthogonality, so the memory misses only the product of the two omitted parts.

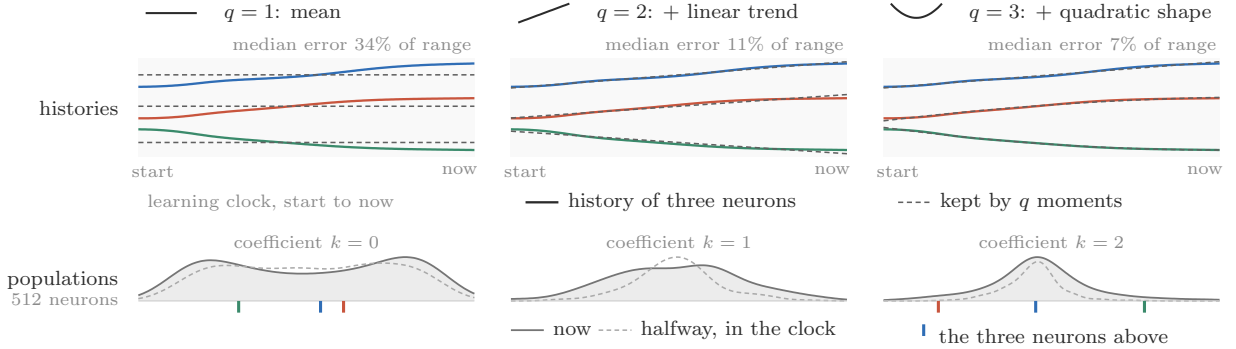


Figure 8: What Legendre moments retain of a history. A network with two hidden tanh layers of width 512 is trained by gradient descent on eight circle points, and the forward responses h of one training input are recorded. Top: three neurons over the whole of training in the learning-speed clock, with their projections onto $q = 1, 2, 3$ Legendre modes (mean, linear trend, quadratic shape). These are projections fitted afterwards to the recorded dense histories, not trajectories of the closure. Above each panel, the median over all 512 neurons of the RMS projection error divided by the range of that neuron's history; for the backward histories $r\delta/\rho$ the same numbers are 26%, 12% and 10%. Bottom: the plotted quantities are the normalized Legendre coefficients $(2k+1)\bar{h}_k/\tau$ of the fitted projection, not the stored moments $\bar{h}_k$; across neurons each coefficient is a population, shown for the history up to the midpoint of the learning clock and for the whole history, with the three neurons above marked.

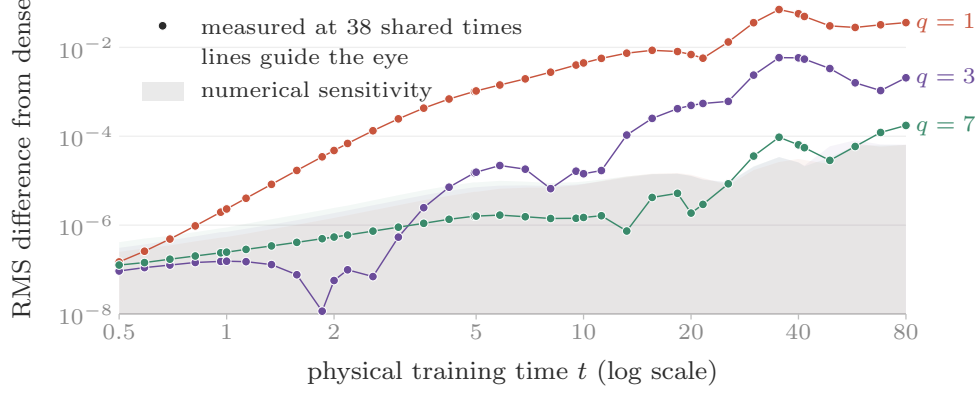


Figure 9: Discrepancy from dense training at common physical times. The paired-label task of Figure 1: RMS difference between response memory and the finer dense replay on 2048 circle queries at the 38 positive shared times, on a logarithmic time axis; the common initialization at $t = 0$ is omitted. Dots are measured; thin lines guide the eye. Light shading extends from the axis floor to each order’s coarse/fine sensitivity scale, obtained by summing the dense and closure prediction changes between the two integration tolerances; it is a numerical diagnostic, not a confidence interval or a certified error bound, and the smallest discrepancies of $q = 3$ and $q = 7$ lie at that scale.

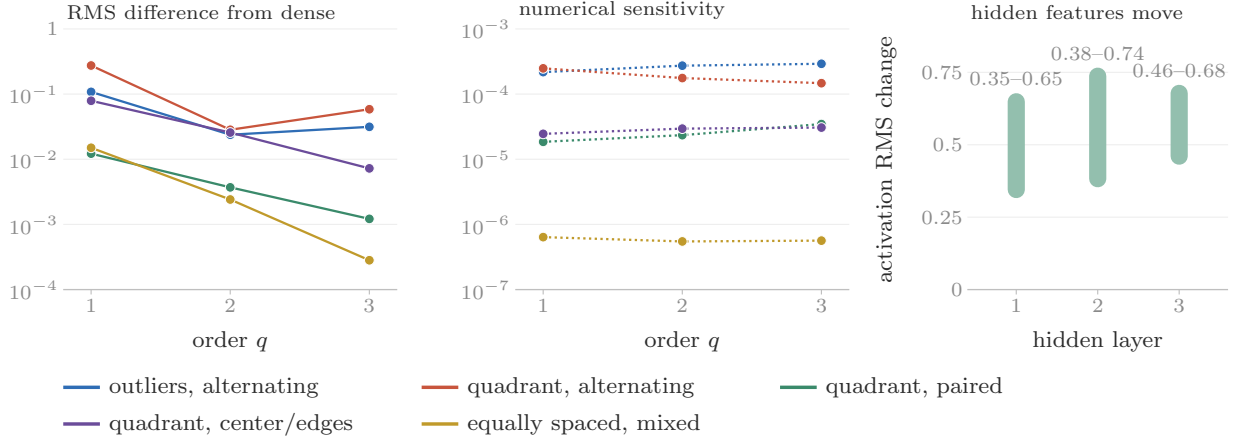


Figure 10: Measured order trends and substantial feature movement. Three-hidden-layer circle experiments, width 4096, at individually fitted MSE 0.001 endpoints. Left: discrepancies on 8192 angles. Middle: sums of dense and closure coarse/fine prediction RMS changes, shown on a separate scale; these are numerical diagnostics, not confidence intervals or certified bounds. Right: ranges of absolute training-activation RMS changes from initialization across the 20 finest runs. Neither the endpoint comparisons nor the three tested orders establish a temporal tracking exponent.

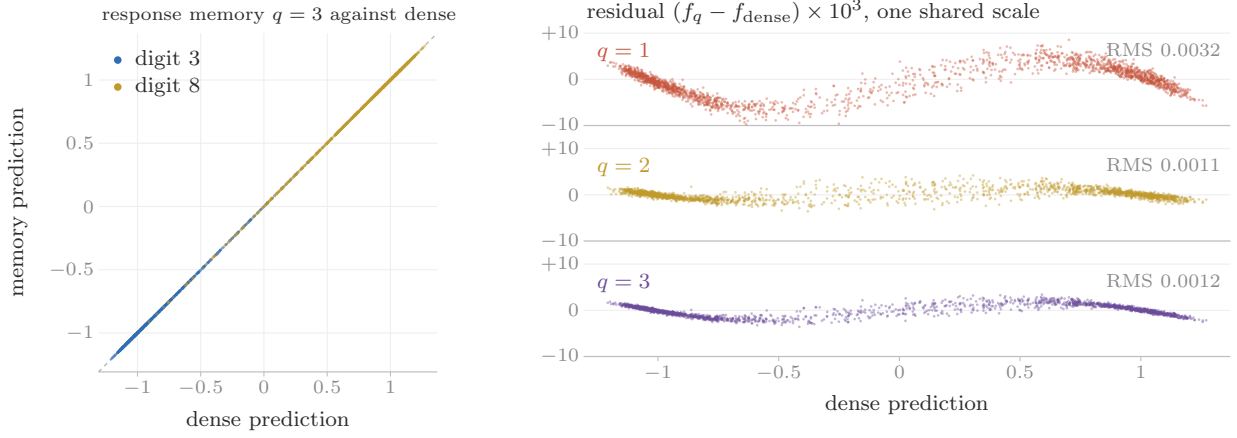


Figure 11: Agreement and its residual structure on MNIST. Digits 3 versus 8, 100 training images, two hidden tanh layers of width 4096. Left: memory $q = 3$ versus dense predictions on all 1984 held-out images. Right: signed differences $f_q - f_{\text{dense}}$, displayed in units of 10^{-3} with one vertical scale for all orders. RMS differences are 0.003248, 0.001084 and 0.001199. Each model is evaluated at its own training-MSE 0.001 endpoint. These are function discrepancies, not label RMSE.

J Frozen-kernel and same-rank controls

The frozen control uses the empirical tangent kernel of the same initialized finite network, including the learning-rate metric. If $J_0(x)$ is its row parameter Jacobian and D has block mobilities $(n, 1, n)$, then $K_0(x, x') = J_0(x)DJ_0(x')^\top$. For the training list X and unhalved mean squared loss, $\dot{f}(X, t) = -(2/m)K_0(X, X)(f(X, t) - y)$. Writing $K_0(X, X) = V \text{diag}(\lambda_j)V^\top$, the test prediction is

$$f_{\text{NTK}}(x, t) = f_0(x) + K_0(x, X)V \text{diag}\left(\frac{1 - e^{-2\lambda_j t/m}}{\lambda_j}\right)V^\top(y - f_0(X)).$$

We evaluate this expression directly and locate the training-MSE 0.001 crossing by bisection, without ridge regularization or discarded eigenmodes. For the equally spaced odd task, the four-representative antipodal quotient removes exact redundancy and preserves the original eight-input mean loss. Kernel blocks are checked against automatic differentiation, and the spectral solution against a matrix exponential and training cross-kernel predictions. The center/edges kernel is ill-conditioned (condition number 2.29×10^9); the two training-prediction formulas agree within 2.1×10^{-8} .

Figures 12 and 13 retain the compact and detailed comparisons with direct factor training. Figure 14 adds the frozen kernel on all five tasks, keeping $q = 3$ and both factor seeds throughout. Memory is closest to the dense endpoint in every row. These are fitted-function comparisons at individual stopping times; the kernel is that of this feature-learning parameterization, rather than a separately scaled infinite-width lazy model. All source predictions, kernel calculations, and reproduction commands are retained with the figure scripts.

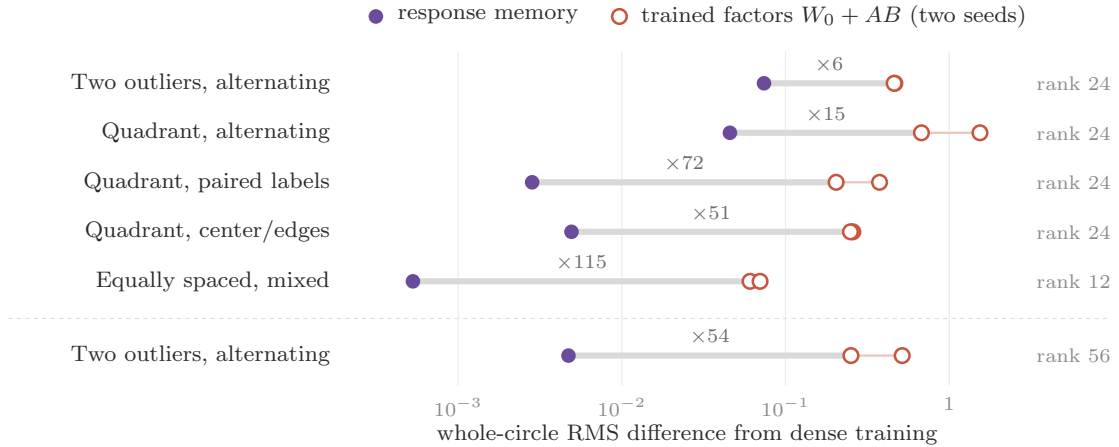


Figure 12: Same rank, different dynamics. Whole-circle RMS difference from dense training for response memory and for a directly trained correction $W_0 + AB$ of the middle matrix with the same rank (two factor seeds), two hidden tanh layers, width 2048. Rank 24 corresponds to $q = 3$, rank 12 to $q = 3$ on the four-representative antipodal quotient, and rank 56 to $q = 7$; labels give the smaller of the two ratios. Matched-loss endpoint comparisons against this specific factor baseline, not against every low-rank method; all ranks and the fitted functions are in Figure 13.

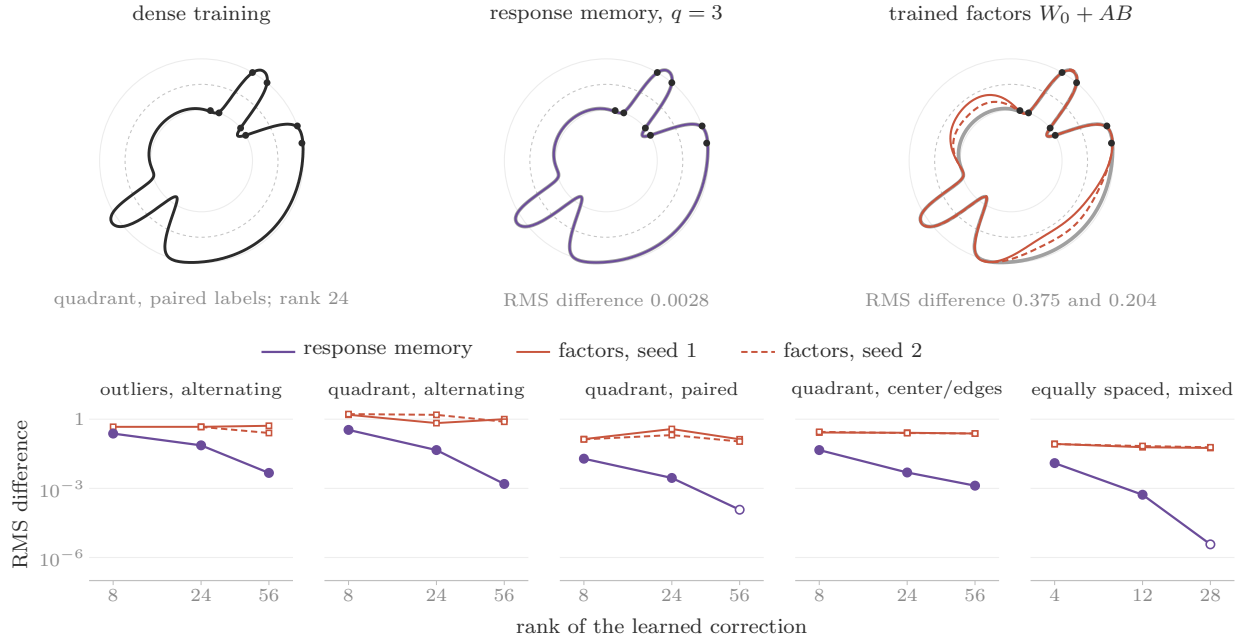
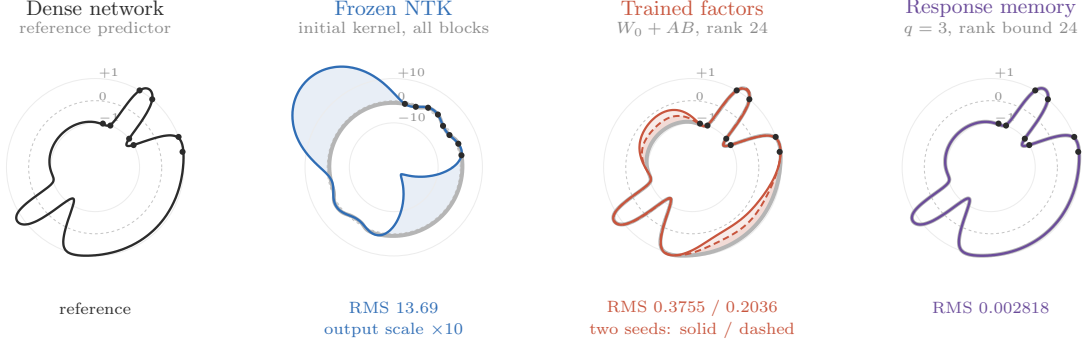
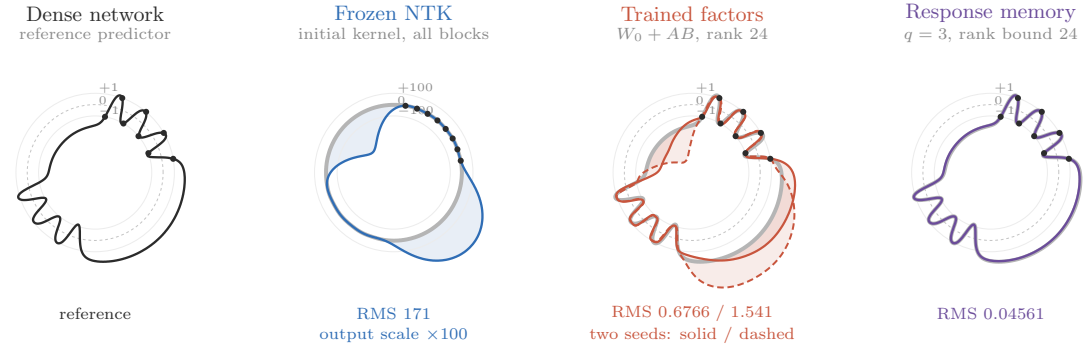


Figure 13: Same correction rank, different fitted functions. Two hidden tanh layers of width 2048. Top: paired-label task, memory $q = 3$ and both factor seeds at rank 24, with the dense reference in black. Bottom: all five tasks at rank bounds 8, 24, 56 (or 4, 12, 28 for the exact antipodal quotient). Every model is shown at its own MSE 0.001 endpoint; RMS uses 8192 angles. Factors use $W_2 = W_0 + AB$, $A(0) = 0$, independent $B_{ij}(0) \sim N(0, 1/\text{rank})$, and unit factor mobilities. Both outer layers have their canonical mobilities. Equal rank matches the factor-array budget, not total state or runtime. One capped comparison is omitted; hollow memory points have limited numerical precision.

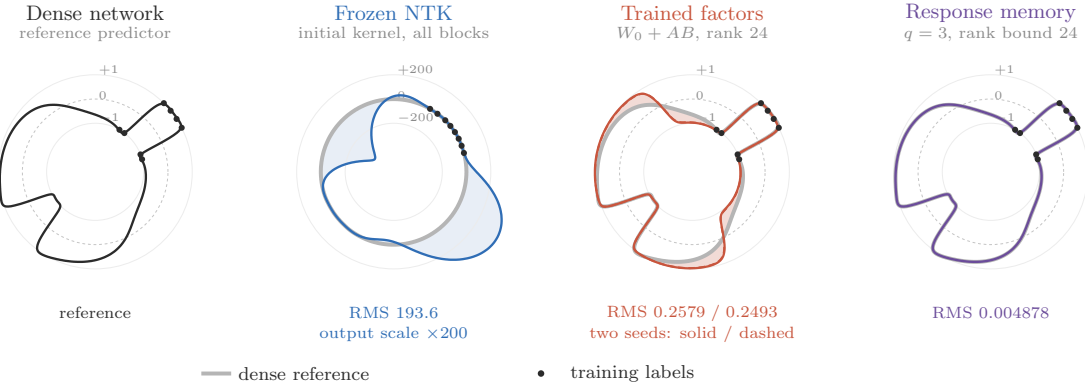
Quadrant, paired labels



Quadrant, alternating



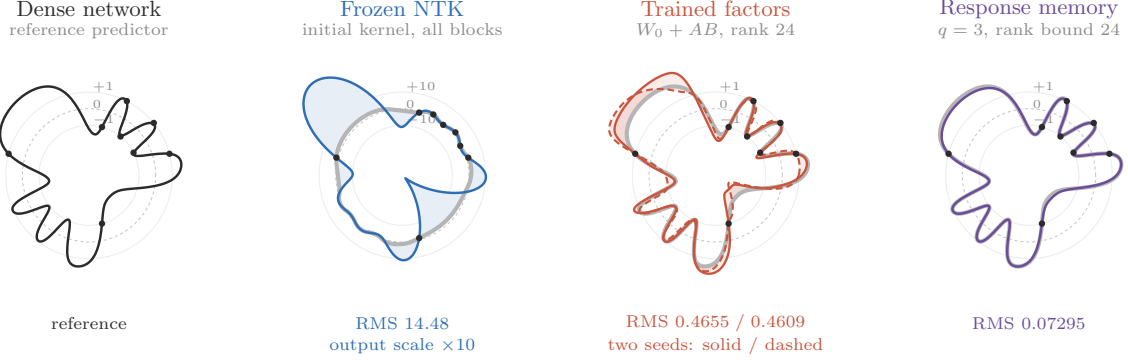
Quadrant, center/edges



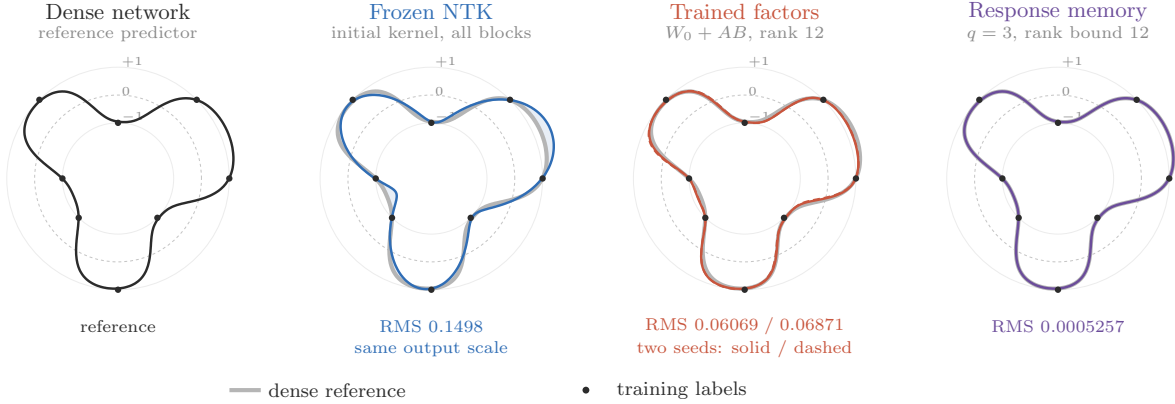
Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 14: Four training descriptions on five circle tasks, part one. Paired-label, alternating-label, and center/edges quadrant tasks. Columns show dense, frozen NTK, both directly trained factor seeds, and memory $q = 3$. The latter two have correction rank bound 24. Each model is at its own training-MSE 0.001 endpoint. The NTK output scale is wider by the marked factor; signed ticks and RMS retain original output units. Gray is the dense reference, black dots are training labels, and RMS uses 8192 circle queries. The next page continues with the remaining two tasks.

Two outliers, alternating



Equally spaced, mixed



Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 14: Four training descriptions on five circle tasks, continued. Two-outlier alternating labels and equally spaced mixed labels. The latter uses four antipodal representatives and correction rank 12; the former uses rank 24. All eight physical training inputs are displayed. NTK uses a ten-times wider output scale for the outlier task and the same scale as the other methods for the equally spaced task. Models, endpoints, seeds, colors and RMS definitions follow the preceding page.